# POINT SET TOPOLOGY

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The following notes were taken when I was studying point-set topology in the summer of 2023. The textbook used was Munkres's *Topology*.

These notes may contain mistakes. Some of the exercises and statements are not provided with a proof. Please use them with caution and verify the information as needed.

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# Contents

## 12 Topological Spaces

**Definition 12.1.** *A* ***topology*** *on a set $X$ is a collection $\mathcal{T}$ of subsets of $X$ having the following properties:*

*(i)* $\varnothing$ *and* $X \in \mathcal{T}$.

*(ii) For any* $U_1, ... \in \mathcal{T}, \bigcup_{i=1}^{\infty} U_i \in \mathcal{T}$.

*(iii) For any* $U_1, ..., U_k \in \mathcal{T}, \bigcap_{i=1}^{k} U_i \in \mathcal{T}$.

*A set $X$ with a defined topology $\mathcal{T}$ is called a* ***topological space****.*

**Definition 12.2.** *$U$ is called an* ***open set*** *of $X$ if $U \in \mathcal{T}$.*

**Definition 12.3.** *The* ***discrete topology*** *on a set $X$ is a topology $\mathcal{T}$ s.t.*

$$\forall U \subset X, U \in \mathcal{T}.$$

*The* ***indiscrete topology*** *(or the trivial topology) on $X$ is $\mathcal{T} = \{\varnothing, X\}$.*

**Definition 12.4.** *Let $X$ be a set. The* ***finite complement topology*** *is $\mathcal{T}_f = \{U \subset X \mid X - U$ finite or $U = \emptyset\}$. The* ***countable complement topology*** *$\mathcal{T}_c$ is defined analogously.*

**Definition 12.5.** *Let $\mathcal{T}$ and $\mathcal{T}'$ be two topologies on a given set $X$. If $\mathcal{T} \subset \mathcal{T}'$, then $\mathcal{T}'$ is said to be* ***finer*** *than $\mathcal{T}$ and $\mathcal{T}$ is said to be* ***coarser*** *than $\mathcal{T}'$. $\mathcal{T}$ and $\mathcal{T}'$ are* ***comparable*** *with each other if either $\mathcal{T} \subset \mathcal{T}'$ or $\mathcal{T} \supset \mathcal{T}'$.*

## 13 Basis for a Topology

**Definition 13.1.** *A **basis** for a topology on $X$ is a collection of $U \subset X$, $\mathcal{B}$, s.t.*

*(i) For any $x \in X, \exists B \in \mathcal{B}$, s.t. $x \in B$.*

*(ii) If $x \in B_1$ and $x \in B_2$, then $\exists B_3 \in \mathcal{B}$, s.t. $x \in B_3$, and $B_3 \subset B_1 \cap B_2$.*

**Theorem 13.2.** *There is a topology $\mathcal{T}$ generated by $\mathcal{B}$ by the following rule:*

$$U \in \mathcal{T} \text{ iff } \forall x \in U, \exists B \in \mathcal{B} \text{ s.t. } x \in B \text{ and } B \subset U.$$

*Proof.* (i) Apparently, $\varnothing$ and $X$ are open under $\mathcal{T}$.

(ii) Let $\{U_\alpha\}_{\alpha \in J}$ be a family of open subsets of $X$, let $U = \bigcup_{\alpha \in J} U_\alpha$.

Then for any $x \in U$, $\exists \beta \in J$ s.t. $x \in U_\beta$.

By definition of $\mathcal{B}$, $\exists B \in \mathcal{B}$ s.t. $x \in B$ and $B \subset U_\beta$, and therefore $B$ is a subset of $U$, so $U$ is open as desired.

(iii) Let $U_1, U_2, ..., U_k$ be open subsets of $X$, let $U = \bigcap_{i=1}^k U_i$.

For any $x \in U$, we have $x \in U_i, \forall i$.

Therefore, $\exists B_1, ..., B_k$ s.t. $x \in B_i \subset U_i$.

By definition and induction, we have some $B \in \mathcal{B}$ s.t. $x \in B$, $B \subset \bigcap B_i \subset \bigcap U_i = U$, so $U$ is open as desired.

□

**Lemma 13.3.** *Let $\mathcal{B}$ be the basis for a topology $\mathcal{T}$ on $X$, then $\mathcal{T}$ equals the collection of all unions of elements of $\mathcal{B}$.*

*Proof.* (i) Clearly, for any $B \in \mathcal{B}$, $B$ is open, so an arbitrary union of elements of $\mathcal{B}$ is open.

(ii) Let $U$ be an open subset of $X$.

For any $x \in U$, $\exists B_x \in \mathcal{B}$, s.t. $x \in B_x \subset U$.

We then have $U = \bigcup_{x \in U} B_x$ so any open subset is an union of basis elements.

□

The above lemma offers a way for understanding what is contained in a topology $\mathcal{T}$ given the basis $\mathcal{B}$ generating it.

The lemma below, on the other hand, offers a way for finding a basis $\mathcal{B}$ that generates a given topology $\mathcal{T}$.

**Lemma 13.4.** *Suppose $\mathcal{C}$ is a collection of open sets on $X$ s.t. for any open subset $U$ and $x \in U$, $\exists C \in \mathcal{C}$ s.t. $x \in C \subset U$, then $\mathcal{C}$ is a basis for the topology $\mathcal{T}$ on $X$.*

*Proof.* We first prove that $\mathcal{C}$ is a basis for some topology $\mathcal{T}'$ on $X$.

(i) Since $X$ is open, for any $x \in X$, there exists some $C \in \mathcal{C}$ s.t. $x \in C$.

(ii) Notice that for any $C_1$, $C_2 \in \mathcal{C}$, $C_1 \cap C_2$ is also open, so $\exists C \in \mathcal{C}$ with $C \subset C_1 \cap C_2$.

Therefore, $\mathcal{C}$ is a basis for some topology $\mathcal{T}'$. Now we prove that $\mathcal{T} = \mathcal{T}'$.

(i) For any $U \in \mathcal{T}$, and any $x \in U$, we have that $x \in C \subset U$ for some $C_x \in \mathcal{C}_\S$. Therefore, $U = \bigcup_{x \in U} C_x$ is an union of elements in $\mathcal{C}$ and therefore open under $\mathcal{T}'$ as well.

(ii) On the other hand, suppose $V \in \mathcal{T}'$, then $V = \bigcup_{\alpha \in J} C_\alpha$.

Since each of $C_\alpha$ is open under $\mathcal{T}$, $V$ is also open under $\mathcal{T}$.

Therefore, the original topology $\mathcal{T}$ is the same as the topology generated by $\mathcal{C}$. □

**Lemma 13.5.** *Let $\mathcal{B}$ and $\mathcal{B}'$ be the basis for topology $\mathcal{T}$ and $\mathcal{T}'$ on $X$, then the followings are equivalent:*

*(i)* $\mathcal{T} \subset \mathcal{T}'$.
*(ii)* *For any* $x \in X$ *and* $B \in \mathcal{B}$ *s.t.* $x \in B, \exists B' \in \mathcal{B}'$ *s.t.* $x \in B'$ *and* $B' \subset B$.

*Proof.* (i) Suppose that $\mathcal{T} \subset \mathcal{T}'$. We know that any basis element $B \in \mathcal{B}$ is open under $\mathcal{T}$, so it is also open under $\mathcal{T}'$ and by definition ii) is met.

(ii) On the other hand, if ii) is met, then for any $U$ open under $\mathcal{T}$ and $x \in U$, $\exists B \ni x$ with $B \subset U$. By ii), there exists $B' \in \mathcal{B}'$ with $x \in B' \subset B \subset U$, so $U$ is also open under $\mathcal{T}'$ and $\mathcal{T} \subset \mathcal{T}'$.

□

**Definition 13.6.** *Let $\mathcal{B}$ be the set of all open intervals in the real line,*

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}.$$

*then the topology generated by $\mathcal{B}$ is called the **standard topology** on $\mathbb{R}$.*

*Similarly, let $\mathcal{B}'$ be the set of all half-open intervals in the real line,*

$$[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\},$$

*the topology generated by $\mathcal{B}'$ is called the **lower limit topology** on $\mathbb{R}$. Under this topology, $\mathbb{R}$ is denoted as $\mathbb{R}_l$.*

*Finally, let $K = \{\frac{1}{n} \mid n \in \mathbb{N}\}$. Let $\mathcal{B}''$ be the set of all open intervals $(a, b)$ and the intervals of form $(a, b) - K$. The topology generated by $\mathcal{B}''$ is called the **K-topology** on $\mathbb{R}$. Under this topology, $\mathbb{R}$ is denoted as $\mathbb{R}_K$.*

**Lemma 13.7.** *The topologies of $\mathbb{R}_l$ and $\mathbb{R}_K$ are strictly finer than the standard topology on $\mathbb{R}$, but are not comparable to each other.*

*Proof.* (i) For any $(a, b) \in \mathcal{B}$ and $x \in (a, b)$, $[x, b) \in \mathcal{B}' \subset \mathcal{B}$, and $x \in [x, b)$ is not in any basis element $(a, c)$ that is a subset of $[x, b)$.

Therefore, the topology on $\mathbb{R}_l$ is strictly finer than the standard topology on $\mathbb{R}$.

(ii) Clearly the topology of $\mathbb{R}_k$ is finer than the standard topology on $\mathbb{R}$, since $\mathcal{B} \subset \mathcal{B}''$.

On the other hand, $0 \in (-1, 1) - K$ is not in any basis element $(a, b)$ that is a subset of $(-1, 1) - K$.

Therefore, the topology on $\mathbb{R}_K$ is strictly finer than the standard topology on $\mathbb{R}$.

(iii) $0 \in [0, 1)$ is not in any basis element of $\mathbb{R}_K$ that is a subset of $[0, 1)$.

On the other hand, $0 \in (-1, 1) - K$ is not in any basis element $[a, b)$ that is a subset of $(-1, 1) - K$.

Therefore, the topologies on $\mathbb{R}_l$ and $\mathbb{R}_K$ are not comparable.

□

**Definition 13.8.** *A **subbasis** $\mathcal{S}$ for a topology on $X$ is a collection of subsets of $X$ whose union equals $X$. The **topology generated by** $\mathcal{S}$ is the collection $\mathcal{T}$ of all unions of finite intersections of elements of $\mathcal{S}$.*

It is not difficult to verify that the topology generated by a subbasis $\mathcal{S}$ is indeed a topology by checking that $\mathcal{B}$, the set of all finite intersections of elements in $\mathcal{S}$, form a basis of $X$.

## Exercises

**Problem 8.** *(i) Show that the countable collection*

$$\mathcal{B} = \{(a, b) \mid a < b, a\ , b \in \mathbb{Q}\}$$

*is a basis that generates the standard topology on $\mathbb{R}$.*

*(ii) Show that the collection*

$$\mathcal{C} = \{[a, b) \mid a < b, a\ , b \in \mathbb{Q}\}$$

*is a basis that generates a different topology from the topology of $\mathbb{R}_l$.*

*Proof.* (i) It suffices to show that for any $(a,b) \subset \mathbb{R}, x \in (a,b)$, there exists $(c,d)$ with $c \in \mathbb{Q}, d \in \mathbb{Q}$ s.t. $x \in (c,d), a < c, b < d$.

Let $c$ be the rational number between $a$ and $x$, $d$ be the rational number between $x$ and $b$, then the condition is satisfied.

(ii) Consider $[\sqrt{2}, 2)$ open in $\mathbb{R}_l$, there does not exist $[a,b)$ s.t. $a \in \mathbb{Q}, b \in \mathbb{Q}, \sqrt{2} \in [a,b)$, and $[a,b) \subset [\sqrt{2}, 2)$.

Thus, the two topologies are different.

$\square$

## 14 Order Topology

**Definition 14.1.** *Let $X$ be a set with a simple order relation $<$. The* ***intervals*** *of $X$ determined by two elements $a, b \in X, a < b$ are*

$$\begin{aligned}(a,b) &= \{x \in X \mid a < x < b\},\\ [a,b) &= \{x \in X \mid a \le x < b\},\\ (a,b] &= \{x \in X \mid a < x \le b\},\\ [a,b] &= \{x \in X \mid a \le x \le b\}.\end{aligned}$$

*The first type is called* ***open intervals****, the last type is called* ***closed intervals****, and the other two types are called* ***half-open intervals****.*

**Definition 14.2.** *Let $X$ be a set with a simple order relation $<$ with more than one element. Let $\mathcal{B}$ be the set of intervals of $X$ in one of the forms*

$$\begin{gathered}(a,b), a, b \in X,\\ [a_0, b), a_0 \text{ the smallest element in } X,\ b \in X,\\ (a, b_0], b_0 \text{ the largest element in } X,\ a \in X.\end{gathered}$$

*Then $\mathcal{B}$ forms a basis for a topology on $X$. The topology is called the* ***order topology.***

It is not difficult to verify that $\mathcal{B}$ forms a basis. An open set would thus be the union of intervals in the three forms.

**Definition 14.3.** *Let $X$ be an ordered set, $a \in X$. The* ***rays*** *of $X$ determined by $a$ are intervals of $X$ in one the forms*

$$\begin{aligned}(a,+\infty) &= \{x \in X \mid x > a\},\\ [a,+\infty) &= \{x \in X \mid x \ge a\},\\ (-\infty,a) &= \{x \in X \mid x < a\},\\ (-\infty,a] &= \{x \in X \mid x \le a\}.\end{aligned}$$

*The first and third type are called* ***open rays****, while the second and the fourth type are called* ***closed rays****.*

**Corollary 14.4.** *The open rays form a subbasis for the order topology of $X$.*

*Proof.* (i) Open rays are open because, if there exists a largest element $b \in X$, then $(a,+\infty) = (a,b]$, which is a basis element for the order topology and is thus open.

On the other hand, if there does not exist a largest element, then

$$(a,+\infty) = \bigcup_{b \in X \mid a < b} (a,b),$$

which is open in the order topology.

Therefore, the topology generated by the open rays as a subbasis is contained in the order topology.

(ii) On the other hand, any element in the basis for the order topology is itself an intersection of finitely many open rays, since $(a,b) = (a,+\infty) \cap (-\infty, b)$, and $(a, b_0]$ and $[a_0, b)$ are open rays themselves.

Thus, the open rays form a subbasis for the topology on $X$.

□

## 15 The Product Topology on $X \times Y$

**Definition 15.1.** *Let $X$ and $Y$ be topological spaces. The **product topology** on $X \times Y$ is the topology with the basis*

$$\mathcal{B} = \{(U \times V) \mid U,\ V \text{ open in } X,\ Y\}.$$

It is not difficult to see that this is a well-defined basis.

**Theorem 15.2.** *If $\mathcal{B}$ is a basis for a topology on $X$ and $\mathcal{C}$ is a basis for a topology on $Y$ then*

$$\mathcal{D} = \{B \times C \mid B \in \mathcal{B},\ C \in \mathcal{C}\}$$

*is a basis that generates the product topology on $X \times Y$.*

*Proof.* We apply **Lemma 13.4**. Let $W \subset X \times Y$ be open. For any $x \times y \in W$, we have some $U \times V \subset X \times Y$ where $U$, $V$ are open in $X$, $Y$, respectively, s.t. $x \times y \in U \times V \subset W$.

Since $\mathcal{B}$ and $\mathcal{C}$ are bases for $X$ and $Y$, $\exists B \in \mathcal{B}$ and $C \in \mathcal{C}$ s.t. $x \in B \subset U$, $y \in C \subset V$. Thus, $x \times y \in B \times C \subset U \times V \subset W$.

Therefore, $\mathcal{D}$ is a basis for the product topology on $X \times Y$. □

**Remark 15.3.** *Notice that in the above proof, directly showing that $\exists B \in \mathcal{B}$ such that $x \in B$ and $B \subset \pi_1(W)$ does not work because $\pi_1(W)$ might not be open in $X$ (so this may not hold). An example for this would be $D^1 \subset \mathbb{R}^2$. It is not the product of any open intervals of $\mathbb{R}$.*

**Definition 15.4.** *We can now define the **standard topology** on $\mathbb{R}^2$ as the product of the order topology on $\mathbb{R}$.*

*The above theorem tells us that subsets of $\mathbb{R}^2$ in the form $(a, b) \times (c, d)$ (i.e., the rectangles) form a basis for this standard topology.*

**Definition 15.5.** *Let $\pi_1 : X \times Y \to X$ be defined by the equation*

$$\pi_1(x, y) = x;$$

*let $\pi_2 : X \times Y \to Y$ be defined by the equation*

$$\pi_2(x, y) = y;$$

*$\pi_1$ and $\pi_2$ are called the **projections** of $X \times Y$ onto its first and second factors.*

**Theorem 15.6.** *$\mathcal{S} = \{\pi_1^{-1}(U) \mid U \text{ open in } Y\} \cup \{\pi_2^{-1}(V) \mid V \text{ open in } Y\}$ forms a subbasis for the product topology of $X \times Y$.*

*Proof.* Let $\mathcal{T}$ be the product topology on $X \times Y$, and $\mathcal{T}'$ be the topology generated by the subbasis $\mathcal{S}$.

Notice that every element of $\mathcal{S}$ is in $\mathcal{T}$ so $\mathcal{T}' \subset \mathcal{T}$.

On the other hand, for any basis element for $\mathcal{T}$, $U \times V$, $U \times V = \pi_1^{-1}(U) \cap \pi_2^{-1}(V)$, so $U \times V \in \mathcal{T}'$, indicating that $\mathcal{T} \subset \mathcal{T}'$.

Therefore, $\mathcal{S}$ is indeed the basis for the product topology on $X \times Y$. □

## 16 The Subspace Topology

**Definition 16.1.** *Let $X$ be a topological space with topology $\mathcal{T}$. If $Y \subset X$, then the set*

$$\mathcal{T}_Y = \{Y \cap U \mid U \in \mathcal{T}\}$$

*is a topology on $Y$, called the* ***subspace topology*** *on $Y$. $Y$ is called a* ***subspace*** *of $X$ with this topology.*

It is not difficult to verify that $\mathcal{T}_Y$ is a topology.

**Lemma 16.2.** *If $\mathcal{B}$ is a basis for the topology of $X$ then*

$$\mathcal{B}_Y = \{B \cap Y \mid B \in \mathcal{B}\}$$

*is a basis for the subspace topology on $Y$.*

*Proof.* Let $U \subset X$ be open and let $y \in U \cap Y$. By the definition of basis, $\exists B \in \mathcal{B}$ s.t. $y \in B \subset U$.

Since $y \in B \cap Y \subset U \cap Y$, $\mathcal{B}_Y$ is thus a well-defined basis for the subspace topology on $Y$. □

**Lemma 16.3.** *Let $Y$ be a subspace of $X$. If $U$ is open in $Y$ and $Y$ is open in $X$, then $U$ is open in $X$.*

*Proof.* By the definition of subspace topology, $U = V \cap Y$ for some $V$ open in $X$.

Since $Y$ and $V$ are open in $X$, we have $Y \cap V = U$ open in $X$. □

**Theorem 16.4.** *If $A$ is a subspace of $X$ and $B$ is a subspace of $Y$, then the product topology on $A \times B$ is the same as the subspace topology of $X \times Y$ on $A \times B$.*

*Proof.* Notice that with subspace topology, the basis elements of $A \times B$ are in the form $U \times V \cap A \times B = (U \cap A) \times (V \cap B)$, where $U$, $V$ are open in $X$, $Y$, respectively.

On the other hand, with product topology, the basis elements of $A \times B$ are also in the form $(U \cap A) \times (V \cap B)$, so the two topologies have the same basis and thus are the same. □

Let $X$ be an ordered set with order topology, $Y$ be a subset of $X$. The inherited order topology on $Y$ need not to be the same as the subspace topology on $Y$.

**Example 16.5.** *Let $X = \mathbb{R}$ and $Y = [0, 1] \cup \{2\}$. $\{2\}$ is open in the subspace topology on $Y$ since $\{2\} = (\frac{3}{2}, \frac{5}{2}) \cap Y$.*

*On the other hand, $2$ is not open in the order topology on $Y$, as all the open sets in $Y$ containing $2$ are in the form $\{x \in Y, a < x \leq 2\}$ for some $a \in Y$.*

**Definition 16.6.** *Given an ordered set $X$, $Y \subset X$ is* ***convex*** *in $X$ if for any $a < b$ in $Y$, $(a, b) \subset Y$.*

**Theorem 16.7.** *Let $X$ be an ordered set in the order topology; let $Y \subset X$ be convex in $X$. Then the order topology on $Y$ is the same as the topology $Y$ inherits as a subspace of $X$.*

*Proof.* Consider the subbasis element in the order topology of $X$, $(a, +\infty)$. By our discussion earlier, $\{(a, +\infty) \cap Y\}$ forms a subbasis of $Y$.

If $a \in Y$, then $(a, +\infty) \cap Y = \{x \in Y \mid x > a\}$, which is an open ray in Y and is open in the order topology on $Y$.

Otherwise, if $a \notin Y$, then $(a, +\infty) \cap Y = Y$ or $\varnothing$. In both case, the intersection is still open in the order topology on $Y$.

Thus, the order topology on $Y$ contains the subspace topology it inherited.

To show the reverse, notice that the open rays in $Y$ is always the intersection of an open ray in $X$ and $Y$. Since the open rays form a subbasis of the order topology on $Y$, the order topology is contained in the subspace topology. □

Proving the statement with the basis element instead of the subbasis element (the open rays) is also valid and analogous to the proof above.

## Exercises

**Problem 3.** *Consider $Y = [-1, 1]$ as a subspace of $\mathbb{R}$, which of the followings are open in $Y$ and which are open in $\mathbb{R}$?*

$$A = \{x \mid \tfrac{1}{2} < |x| < 1\},$$
$$B = \{x \mid \tfrac{1}{2} < |x| \leq 1\},$$
$$C = \{x \mid \tfrac{1}{2} \leq |x| < 1\},$$
$$D = \{x \mid \tfrac{1}{2} \leq |x| \leq 1\},$$
$$E = \{x \mid \{0 < |x| < 1 \text{ and } \tfrac{1}{x} \notin \mathbb{N}\}.$$

*Proof.* $A$ is open in both spaces and $B$ are both open in $Y$. $A$ is a basis element in both topologies (the subspace topology on $Y$ is the same as the order topology on $Y$), while $B = (\frac{1}{2}, 1) \cap Y$ and the former set is open in $\mathbb{R}$.

$C$ is not open in both topologies because there is no basis element $U$ in the order topology of $Y$ s.t. $\frac{1}{2} \in U$ and $U \subset C$. The same reasoning applies for $D$.

$E$ is open in both topologies since $E = \bigcup_{i=1}^{\infty}(\frac{1}{i+1}, \frac{1}{i}) \cup \bigcup_{i=1}^{\infty}(-\frac{1}{i}, -\frac{1}{i+1})$, which is a union of basis elements in both topologies. □

**Problem 7.** *Let $X$ be an ordered set. If $Y$ is a proper subset of $X$ and is convex in $X$, does it follow that $Y$ is an interval or a ray in $X$?*

*Proof.* No. Consider $X = \mathbb{Q}, Y = (\sqrt{2}, 2)$. □

**Problem 8.** *Show that the dictionary order topology on the set $\mathbb{R} \times \mathbb{R}$ is the same as the product topology $\mathbb{R}_d \times \mathbb{R}$, where $\mathbb{R}_d$ is the discrete topology on $\mathbb{R}$. Compare this topology with the standard topology on $\mathbb{R}^2$.*

*Proof.* The basis elements for the product topology on $R_d \times R$ is in the form $x \times (y_1, y_2)$, which are open in the dictionary order topology as it is equivalent to $(x \times y_1, x \times y_2)$.

On the other hand, every open set in the dictionary order topology is in the form $(x_1 \times y_1, x_2 \times y_2)$, which is also open in the product topology $R_d \times R$ as it equals $\{x_1\} \times (y_1, \infty) \cup \{x_2\} \times (-\infty, y_2) \cup \bigcup\limits_{x_1 < x < x_2} \{x\} \times \mathbb{R}$, which is a product of basis elements in the product topology $R_d \times R$.

The dictionary order topology is strictly finer than the standard topology on $\mathbb{R}^2$, as $0 \times (0, 1)$ is open in the former but not the latter. □

## 17 Closed Sets and Limit Points

**Definition 17.1.** *A subset $A$ of a topological space $X$ is said to be **closed** if $X - A$ is open.*

Notice a set can be neither open or closed, or both open and closed.

**Theorem 17.2.** *Let $X$ be a topological space. The followings hold:*

*(i) $\varnothing$ and $X$ are closed.*
*(ii) For any $\{U_i\}$ a family of closed sets, $\bigcap_{i=1}^{\infty} U_i$ is closed.*
*(iii) For any $U_1, ..., U_k$ closed, $\bigcup_{i=1}^{k} U_i$ is closed.*

The theorem is easily proved by applying DeMorgan's Law to what the definition of a topology tells about open sets.

**Theorem 17.3.** *Let $Y$ be a subspace of $X$. A set $A$ is closed in $Y$ iff $X = C \cap Y$, where $C$ is a closed subset of $X$.*

*Proof.* Suppose $A = C \cap Y$, then $Y - A = Y - (X - C) \cap Y$, which is by definition open in $Y$, so $A$ is closed in $Y$.

On the other hand, by definition, $A$ closed in $Y$ is equivalent to $Y - A$ open in $Y$. By the definition of subspace topology, $Y - A$ open is equivalent to $Y - A = U \cap Y$, where $U$ is open in $X$.

This is equivalent to $X - U = C$ being closed in $X$, implying that $A = (X - U) \cap Y = C \cap Y$. □

**Theorem 17.4.** *Let $Y$ be a subspace of $X$. If $A$ is closed in $Y$ and $Y$ is closed in $X$ then $A$ is closed in $X$.*

*Proof.* By the above theorem, $A$ closed in $Y$ so $A = C \cap Y$, where $C$ closed in $X$.

Thus, $A$ is an intersection of two closed sets in $X$ so it is also closed in $X$. □

**Definition 17.5.** *Let $A$ be a subset of a topological space $X$. The **interior** of $A$, Int $A$, is the union of all open sets contained in $A$.*

*The **closure** of $A$, $\bar{A}$, is the intersection of all closed sets containing $A$.*

**Corollary 17.6.** *Int $A \subset A \subset \bar{A}$. If $A$ is open, Int $A = A$; if $A$ is closed, $\bar{A} = A$.*

**Theorem 17.7.** *Let $Y$ be a subspace of $X$, $A \subset Y$. Let $\bar{A}$ be the closure of $A$ in $X$, then the closure of $A$ in $Y$ is $\bar{A} \cap Y$.*

*Proof.*

Let $B$ be the closure of $A$ in $Y$. $\bar{A}$ is closed in $X$ so $\bar{A} \cap Y$ is closed in $Y$, so $\bar{A} \cap Y \supset B$.

On the other hand, we know that $B$ is closed in $Y$, so $B = C \cap Y$, where $C$ is closed in $X$.

Therefore, as $\bar{A} \subset C$ we have that $\bar{A} \cap Y \subset C \cap Y = B$. Thus, $\bar{A} \cap Y = B$. □

**Definition 17.8.** *A set $A$ is said to **intersect** a set $B$ if $A \cap B \neq \varnothing$.*

**Definition 17.9.** *$U$ is a **neighborhood** of $x$ if $U$ is an open set containing $x$.*

**Theorem 17.10.** *Let $A$ be a subset of a topological space $X$.*

*i) $x \in \bar{A}$ iff every neighborhood $U$ of $x$ intersects $A$.*
*ii) $x \in \bar{A}$ iff every basis element $B$ containing $x$ intersects $A$.*

*Proof.* (i) Notice that this statement is equivalent to saying $x \notin \bar{A}$ iff there is a neighborhood $U$ of $x$ s.t. $U \cap A = \varnothing$.

If $x \notin \bar{A}$, then $X - \bar{A}$ is a neighborhood of $x$ disjoint from $A$.

On the other hand, if there is a neighborhood $U$ of $x$ s.t. $U \cap A = \varnothing$, then $X - U$ is a closed set containing $A$, which also contains $\bar{A}$ by definition, therefore, $x \notin \bar{A}$.

(ii) If every basis element $B$ containing $x$ intersects $A$, then so is every neighborhood $U$ of $x$, so $x \in \bar{A}$. The converse is trivial.

□

**Definition 17.11.** *If $A$ is a subset of a topological space $X$ and $x \in A$, then $x$ is a **limit point** of $A$ if every neighborhood of $x$ intersects $A$ at some point other than $x$.*

**Theorem 17.12.** *Let $A$ be a subset of a topological space $X$; let $A'$ be the set of all limit points of $A$. Then*

$$\bar{A} = A \cup A'.$$

*Proof.* If $x \in A'$ then every neighborhood of $x$ intersects $A$, so $x \in \bar{A}$ by the theorem earlier.

On the other hand, if $x \in \bar{A}$, then every neighborhood of $x$ intersects $A$, so either $x$ is in $A$ or there are some other points in $A$ that are in every neighborhood of $x$, indicating that $x \in A'$. □

**Corollary 17.13.** *A subset of a topological space is closed iff it contains all its limit points.*

In some topological spaces the one-point set may not be closed, and a sequence might converge to more than one point (where converging to $x$ of $X$ means for any neighborhood $U$ of $x$, $\exists N \in \mathbb{N}$ s.t. $x_n \in U$ if $n \geq N$.)

To rule these possibilities out, we have:

**Definition 17.14.** *A topological space $X$ is called a **Hausdorff Space** if for any $x_1, x_2 \in X$, $\exists$ neighborhoods $U_1, U_2$ of $x_1, x_2$, respectively, that are disjoint.*

The next theorem is actually weaker than the Hausdorff condition. For example, in the finite complement topology, every finite point sets in $\mathbb{R}$ is closed but $\mathbb{R}$ with the topology is not Hausdorff, as the complements $X - U_1$, $X - U_2$ are both finite so $U_1$ is not a subset of $X - U_2$ and the two sets cannot be disjoint.

The condition such that every finite point set is closed is actually named the ***T1-Axiom***.

**Theorem 17.15.** *Every finite point set in a Hausdorff space $X$ is closed.*

*Proof.* Clearly, every one point set in a Hausdorff space is closed, as if $x \neq x_0$, then $\exists$ neighborhood $U, V$ of $x_0, x$, s.t. $U \cap V = \varnothing$, so $x$ is not a limit point of $x_0$.

Then, any finite point set is a finite union of closed one-point sets so it is closed. □

**Theorem 17.16.** *Let $X$ be a topological space satisfying the T1-Axiom, $A \subset X$. Then the point $x$ is a limit point of $A$ iff every neighborhood of $x$ contains infinitely many points of $A$.*

*Proof.* The if direction is trivial.

Suppose $x$ is a limit point of $A$ with some neighborhood $U$ intersecting $A$ at only finitely many points $\{x_1, ..., x_k\}$.

$X - \{x_1, ..., x_k\}$ is open since $\{x_1, ..., x_k\}$ is closed. Thus, $U \cap X - \{x_1, ..., x_k\}$ is also open and is a neighborhood of $x$ that does not intersect $A$, leading to a contradiction. □

**Theorem 17.17.** *If $X$ is a Hausdorff space, then a sequence of points of $X$ converges to at most one point of $X$.*

*Proof.* Suppose the sequence converges to $x$ and $y$. Since $X$ is Hausdorff, $\exists$ disjoint neighborhood $U, V$ of $x, y$.

If $U$ contains all but finitely many points of the sequence, then $V$ contains at most finitely many points of the sequence, so the sequence does not converge to $y$, leading to a contradiction. □

**Theorem 17.18.** *(i) Every simply ordered set is a Hausdorff space in the order topology.*

*(ii) The product of two Hausdorff space is a Hausdorff space.*

*(iii) A subspace of a Hausdorff space is a Hausdorff space.*

## Exercises

**Problem 9.** *Prove that $\bar{A \times B} = \bar{A} \times \bar{B}$ in the space $X \times Y$ if $A \subset X$ and $B \subset Y$.*

*Proof.* As $\bar{A} \times \bar{B}$ is closed in $X \times Y$ and contains $A \times B$, it contains $\overline{A \times B}$. by definition

On the other hand, suppose $x \times y \in \bar{A} \times \bar{B}$. Let $U$ be a neighborhood of $x$ in $X$, $V$ be a neighborhood of $y$ in $Y$. $U \cap A \neq \varnothing$ and is open in $A$, $V \cap B \neq \varnothing$ and is open in $B$. Thus, $(U \times V) \cap (A \times B) = (U \cap A) \times (V \cap B) \neq \varnothing$ and is a basis element in $A \times B$. Thus, every basis element containing $x \times y$ intersects $A \times B$, so $\bar{A} \times \bar{B} \subset \bar{A \times B}$. □

**Problem 13.** *Prove that $X$ is Hausdorff iff the* ***diagonal*** $\Delta = \{x \times x \mid x \in X\}$ *is closed in $X \times X$.*

*Proof.* Suppose that $X$ is Hausdorff. Then $\forall x, y \in X$, $\exists U, V$ neighborhood of $x, y$ s.t. $U \cap V = \varnothing$. Thus, $x \times y \in U \times V \subset X \times X \backslash \Delta$, so, $X \times X \backslash \Delta$ is open, so $\Delta$ is closed.

On the other hand, suppose that $\Delta$ is closed in $X \times X$. Let $x, y \in X$, $x \neq y$. Then $x \times y \in X \times X \backslash \Delta$ which is open, so $\exists U \times V \subset X \times X \backslash \Delta$ s.t. $U, V$ are open in $X$ and are disjoint, with $x \in U$, $y \in V$. Therefore, $X$ is Hausdorff.

□

**Problem 14.** *In the finite complement topology on $\mathbb{R}$, to what points does the sequence $x_n = 1/n$ converge to?*

*Proof.* Let $x \in R$. For any neighborhood $U$ of $x$, $X \backslash U$ is finite. Thus, there are infinitely many points in $x_n$ that are also in $U$. As a result, $x_n$ converge to all $x \in \mathbb{R}$.

In fact, any sequence with a infinite range converges to any point in $\mathbb{R}$ under the finite complement topology. □

**Problem 19.** *If $A \subset X$, the* ***boundary*** *of $A$ is given by the equation*

$$Bd\ A = \bar{A} \cap \overline{X - A}$$

*(i) Show that Int $A$ and Bd $A$ are disjoint, and $\bar{A} =$ Int $A$ $\cup$Bd $A$.*
*(ii) Show that Bd $A = \varnothing$ iff $A$ is both open and closed.*
*(iii) Show that $U$ is open iff Bd $U = \bar{U} - U$.*
*(iv) If $U$ is open, is it true that $U =$ Int $(\bar{U})$?*

*Proof.*
(i) If $x \in \text{Int } A \cap \text{Bd } A$, then there is a neighborhood $U$ of $x$ that is entirely in $A$ but also intersecting $X - A$, leading to a contradiction.

Notice that if $x \in \text{Bd } A$, then $x \in \bar{A}$ so $\text{Int } A \cup \text{Bd } A \subset \bar{A}$.

On the other hand, if $x \in \bar{A}$, then either all of its neighborhoods intersect $X - A$ or some of its neighborhoods lie entirely in $A$, so $x \in \text{Int } A \cup \text{Bd } A$.

(ii) The if direction is trivial. If $\text{Bd } A = \varnothing$, then $\bar{A} = \text{Int } A$, so $A = \bar{A} = \text{Int } A$, so $A$ is both open and closed.

(iii) $U$ is open iff $X - U$ is closed, which happens iff $\overline{X - U} = X - U$, which happens iff $\text{Bd } U = \bar{U} \cap \overline{X - U} = \bar{U} \cap X - U = \bar{U} - U$.

(iv) No. Consider $U = \mathbb{R} - 0$ in the standard topology of $\mathbb{R}$.

□

## 18 Continuous Functions

**Definition 18.1.** *A function $f : X \to Y$ is **continuous** if for any $V \subset Y$ open in $Y$, $f^{-1}(V)$ is open in $X$.*

**Corollary 18.2.** *A function $f : X \to Y$ is continuous iff $f^{-1}(B)$ is open for any basis element $B$ of $Y$. In fact, $f$ is continuous iff $f^{-1}(S)$ is open for any subbasis element $S$.*

**Theorem 18.3.** *Let $X$ and $Y$ be topological spaces, $f : X \to Y$. The followings are equivalent:*

*(i) $f$ is continuous.*
*(ii) For any $A \subset X$, $f(\bar{A}) \subset \overline{f(A)}$.*
*(iii) For any $B \subset Y$ closed, $f^{-1}(B)$ is closed in $X$.*
*(iv) For any $x \in X$ and neighborhood $V$ of $f(x)$, $\exists$ neighborhood $U$ of $x$ s.t. $f(U) \subset V$.*

*Proof.* (i) $\Rightarrow$ (ii): Assume $f$ is continuous. WTS that if $x \in \bar{A}$, then $f(x) \in \overline{f(A)}$.

Let $V$ be a neighborhood of $f(x)$, then $f^{-1}(V)$ is a neighborhood of $x$. Thus, $f^{-1}(V) \cap A \neq \varnothing$, indicating that there is some $y \in A$ with $y \in f^{-1}(V)$, so $f(y) \in V$ and $f(x) \in \overline{f(A)}$.

(ii) $\Rightarrow$ (iii): Let $B \subset Y$ be closed. We have that $f(\overline{f^{-1}(B)}) \subset \overline{(f(f^{-1}(B)))} \subset \bar{B} = B$.

Therefore, $\overline{f^{-1}(B)} = f^{-1}(B)$ so $f^{-1}B$ is closed.

(iii) $\Rightarrow$ (i): Let $U \subset Y$ be open. We then have that $Y - U$ is closed and $f^{-1}(Y - U) = X - f^{-1}(U)$ which is closed, so $f^{-1}(U)$ is open. Thus, $f$ is continuous.

(i) $\Rightarrow$ (iv): If $V$ is a neighborhood of $f(x)$ in $Y$, then $U = f^{-1}(V)$ is open in $X$ so $U$ is a neighborhood of $x$ such that $f(U) \subset V$.

(iv) $\Rightarrow$ (i): Let $V$ be an open set in $Y$. Then for any $x \in f^{-1}(V)$, $\exists$ neighborhood $U_x$ of $x$ s.t. $f(U_x) \subset V$, so $U_x \subset f^{-1}(V)$.

Therefore, $f^{-1}(V)$ is open as it is the union $\displaystyle\bigcup_{x \in f^{-1}(V)} U_x$. □

**Definition 18.4.** *Let $X$ and $Y$ be topological spaces, with $f : X \to Y$ be bijective. If both $f$ and $f^{-1} : Y \to X$ are continuous, then $f$ is called a **homeomorphism**.*

**Corollary 18.5.** *If $f : X \to Y$ is a homeomorphism, then for any open subset $U$ of $X$, $f(U)$ is open in $Y$.*

**Definition 18.6.** *Let $f : X \to Y$. Let $Z = f(X)$ and take it as a subspace of $Y$. If $f' : X \to Z$ is a homeomorphism then $f$ is called a **topological imbedding** of $X$ into $Y$.*

An example of a bijective continuous function $f : X \to Y$ without being a homeomorphism is:

$$f : [0, 1) \to S^1,$$

where $S^1 = \{x \times y \mid x^2 + y^2 = 1\}$. Let $f(t) = (\cos 2\pi t, \sin 2\pi t)$. Then $f$ is bijective and continuous but $f^{-1}$ is not continuous, as $U = [0, \frac{1}{4})$ is open in $[0, 1)$ while its image is not in $S^1$.

**Theorem 18.7.** *Let $X$, $Y$, and $Z$ be topological spaces. We can construct continuous functions by:*

*(i) If $f : X \to Y$ maps all of $X$ into a single point $y_0$ of $Y$, then $f$ is continuous.*
*(ii) If $A$ is a subspace of $X$, then the inclusion map $j : A \to X$ is continuous.*
*(iii) If $f : X \to Y$ and $g : Y \to Z$ are continuous, then $g \circ f : X \to Z$ is continuous.*
*(iv) If $f : X \to Y$ is continuous, $A$ a subspace of $X$, then $f|_A : A \to Y$ is continuous.*
*(v) Let $f : X \to Y$ be continuous. If $Z$ is a subspace of $Y$ containing the image set $f(X)$, then $g : X \to Z$ by restricting the range of $f$ is continuous.*

*If $Z$ is a space having $Y$ as a subspace, then $h : X \to Z$ by expanding the range of $f$ is continuous.*

*(vi) The map $f : X \to Y$ is continuous if $X$ can be written as the union of open sets $U_\alpha$ s.t. $f|_{U_\alpha}$ is continuous for each $\alpha$.*

*Proof.* (i), (ii), and (iii) are trivial.

(iv) $f|_A$ is the composite of continuous maps $j : A \to X$ and $f$.

(v) $h$ is continuous because it is the composite of $g$ and $f$.

$g$ is continuous because for any $U$ open in $Z$, $U = Z \cap V$ for some $V$ open in $Y$. Therefore, $g^{-1}(U) = f^{-1}(V)$ which is open in $X$.

(vi) Let $V$ open in $Y$. We have that $f^{-1}(V) \cap U_\alpha = (f|U_\alpha)^{-1}(V)$ which is open in $U_\alpha$ and therefore in $X$.

Therefore, $f^{-1}(V) = \bigcup_\alpha (f^{-1}(V) \cap U_\alpha)$, which is also open in $X$, so $f$ is continuous.

□

**Theorem 18.8.** *Let $X = A \cup B$, where $A$ and $B$ are closed in $X$. Let $f : A \to Y$ and $g : B \to Y$ be continuous. If for any $x \in A \cap B, f(x) = g(x)$, then $h : X \to Y$ is continuous by setting $h(x) = f(x)$ if $x \in A$ and $h(x) = g(x)$ if $x \in B$.*

*The same holds if we replace [closed] by [open].*

*Proof.* Let $C$ be closed in $Y$, then $h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$. $f^{-1}(C)$ is closed in $A$ and thus closed in $X$. Similarly, $g^{-1}(C)$ is closed in $X$.

Thus, $h^{-1}(X)$ is closed in $X$ and $h$ is thus continuous. □

**Theorem 18.9.** *Let $f : A \to X \times Y$ be given by*

$$f(a) = (f_1(a), f_2(a)).$$

*Then $f$ is continuous iff $f_1, f_2$ are continuous.*

*Proof.* (i) Let $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ be projections.

If $f$ is continuous, then $f_i = \pi_i \circ f$ is continuous.

(ii) On the other hand, suppose that $f_1$, $f_2$ are continuous.

For any basis element $U \times V \in X \times Y$, $f^{-1}(U \times V) = f_1^{-1}(U) \cap f_2^{-1}(V)$ is open.

Therefore, $f^{-1}((U) \times V)$ is open and $f$ is continuous.

□

## Exercises

**Problem 2.** *Suppose that $f : X \to Y$ is continuous. If $x$ is a limit point of $A \subset X$, then is $f(x)$ always a limit point of $f(A)$?*

*Proof.* No. Consider $f : \mathbb{R} \to x_0$, then $\forall x \in \mathbb{R}$, $x$ is a limit point of $\mathbb{R}$ but $x_0$ is not a limit point of $x_0$. □

**Problem 6.** *Find a function $f : \mathbb{R} \to \mathbb{R}$ that is continuous at exactly 1 point.*

*Proof.* Consider $f(x) = \begin{cases} x, x \in \mathbb{Q} \\ -x, x \in \mathbb{R} \backslash \mathbb{Q} \end{cases}$.

Then $f(x)$ is continuous at and only at $x = 0$. □

**Problem 9.** *Let $\{A_\alpha\}$ be a collection of subsets of $X$ with $X = \bigcup_\alpha A_\alpha$. Let $f : X \to Y$, suppose that $f|_{A_\alpha}$ is continuous for each $\alpha$.*

*(i) Show that if the collection $\{A_\alpha\}$ is finite and each set $A_\alpha$ is closed then $f$ is continuous.*

*(ii) Find an example where the collection $\{A_\alpha\}$ is countable and each $A_\alpha$ is closed, but $f$ is not continuous.*

*(iii) An indexed family of set $\{A_\alpha\}$ is said to be **locally finite** if each point $x$ of $X$ has a neighborhood that intersects only finitely many $A_\alpha$. Show that if the family $\{A_\alpha\}$ is locally finite and each $A_\alpha$ is closed, then $f$ is continuous.*

*Proof.* (i) True by pasting lemma and induction on $|\{A_\alpha\}|$.

(ii) Consider $f : \mathbb{R} \to \mathbb{R}$ with $A_{\alpha_0} = (-\infty, 0]$ and $A_{\alpha_n} = [\frac{1}{n}, +\infty)$.

Let $f|A_{\alpha_0} = 0$ and $f|A_{\alpha_n} = 1$ for $n \in \mathbb{N}$. Then, the function is not continuous at 0.

(iii) For any $x \in X$, let $U_x$ be a neighborhood of it that intersects only finitely many $A_{\alpha_i}$.

Since $A_{\alpha_i}$ is closed, $X - A_{\alpha_i}$ is open, so $U_x \cap X - A_{\alpha_i}$ is open in $U_x$ and $U_x \cap A_{\alpha_i}$ is closed in $U_x$.

Applying (i) on $U_x$ we have that $f|_{U_x}$ is continuous.

Therefore, $f$ is continuous by**Theorem 18.7**.

$\square$

**Problem 11.** *Let $F : X \times Y \to Z$. $F$ is* ***continuous in each variable separately*** *if for any $y_0 \in Y$, the map $h : X \to Z$ defined by $h(x) = F(x \times y_0)$ is continuous. The map $k : Y \to Z$ is defined similarly and is also continuous. Show that if $F$ is continuous then $F$ is continuous in each variable separately.*

*Proof.* $h$ is the composite of two continuous functions $j : X \to X \times \{y_0\}$, $x \to x \times y_0$ and $F$, so $h$ is continuous iff $F$ is. The proof for $k$ is similar. $\square$

**Problem 13.** *Let $A \subset X$; let $f : A \to Y$ be continuous; let $Y$ be Hausdorff. Show that if $f$ may be extended to a continuous function $g : \bar{A} \to Y$, then $g$ is uniquely determined by $f$.*

*Proof.* Suppose there are two continuous functions $g$ and $h$ from $\bar{A}$ to $Y$, and $\exists x \in \bar{A}$ s.t. $g(x) \neq h(x)$, then $x \in A'$.

Since $Y$ is Hausdorff, $\exists U$, $V$ neighborhoods of $g(x)$, $h(x)$, s.t. $U \cap V = \varnothing$. Then $x \in g^{-1}(U) \cap h^{-1}(V)$ so $g^{-1}(U) \cap h^{-1}(V)$ is a neighborhood of $x$.

Since $x \in A'$, $\exists y \in g^{-1}(U) \cap h^{-1}(V) \cap A$ and $y \neq x$.

However, $g(y) = h(y)$, so $g(y) \in U \cap V$, contradicting the assumption. $\square$

## 19 The Product Topology

**Definition 19.1.** *Let $J$ be an index set. Given a set $X$, a **J-tuple** of elements of $X$ is a function $\mathbf{x} : J \to X$.*

*If $\alpha$ is an element of $J$, then the value of $x$ at the $\alpha$-th index is denoted by $x_\alpha$, and $\mathbf{x}$ is denoted by $\{x_\alpha\}_{\alpha \in J}$.*

**Definition 19.2.** *Let $\{X_\alpha\}_{\alpha \in J}$ be an indexed family of topological spaces. Take the basis for the product topology on the space*

$$\prod_{\alpha \in J} X_\alpha$$

*as the collection of all sets of the form*

$$\prod_{\alpha \in J} U_\alpha$$

*where $U_\alpha$ is open in $X_\alpha$, $\forall \alpha \in J$. The resulted topology is called the **box topology**.*

It is not difficult to verify that the above definition gives a well-defined basis.

**Definition 19.3.** *Let $\mathcal{S}_\beta$ be the collection*

$$\mathcal{S}_\beta = \{\pi_\beta^{-1}(U_\beta) \mid U_\beta \text{ open in } X_\beta\}$$

*where $\pi_\beta((x_\alpha)_{\alpha \in J}) = x_\beta$.*
*Then the union of these collections*

$$\mathcal{S} = \bigcup_{\beta \in J} \mathcal{S}_\beta$$

*is a subbasis for a topology on $\prod X_\alpha$. The generated topology is called the **product topology**. $\prod\limits_{\alpha \in J} X_\alpha$ under this topology is called the **product space**.*

**Remark 19.4.** *Let $\mathcal{B}$ be the basis of the product topology, then a typical basis element in $\mathcal{B}$ has the form of*

$$B = \pi_{\beta_1}^{-1}(U_{\beta_1}) \cap \dots \cap \pi_{\beta_n}^{-1}(U_{\beta_n}).$$

*As a result, a typical basis element $B$ is the product $\prod U_\alpha$ where $U_\alpha$ is open in $X_\alpha$ and $U_\alpha \neq X_\alpha$ for finitely many $\alpha$.*

*One thing immediately clear is that the box topology and the product topology are the same on finite Cartesian products $\prod_{a=1}^n X_a$, and the box topology is strictly finer than the product topology if the index set is infinite.*

**Theorem 19.5.** *Suppose the topology on each space $X_\alpha$ is given by the basis $\mathcal{B}_\alpha$. The collection of all sets of the form*

$$\prod_{\alpha \in J} B_\alpha$$

*where $B_\alpha \in \mathcal{B}_\alpha$ will serve as a basis for the box topology on $\prod_{\alpha \in J} X_\alpha$.*

*The collection of all sets of the same form, where $B_\alpha \in \mathcal{B}_\alpha$ for finitely many $\alpha \in J$ and $B_\alpha = X_\alpha$ for other $\alpha$, will serve as a basis for the product topology on $\prod_{\alpha \in J} X_\alpha$.*

*Proof.* Suppose $U = \prod U_\alpha$ is an open set in the box topology. For any $\curvearrowleft \in U$, we have that $x \in B_\alpha \in U_\alpha$ for some basis element $B_\alpha$ of $X_\alpha$, by the definition of basis.

Therefore, $\mathbf{x} \in \prod X_\alpha \in \prod U_\alpha$ so the collection of all sets of the given form is indeed a basis for the box topology.

The reasoning is analogous for the product topology. $\square$

**Example 19.6.** *In $\mathbb{R}^n$, a basis element is in the form*

$$(a_1, b_1) \times \dots \times (a_n, b_n).$$

**Theorem 19.7.** *Let $A_\alpha$ be a subspace of $X_\alpha$, for each $\alpha \in J$. Then $\prod A_\alpha$ is a subspace of $\prod X_\alpha$ if both products are given the box topology or the product topology.*

*Proof.* Notice that basis in $\prod A_\alpha$ in the subspace topology are in the form $\prod U_\alpha \cap \prod A_\alpha = \prod(U_\alpha \cap A_\alpha)$, where $U_\alpha$ is open in $X_\alpha$, which are exactly the basis elements for $\prod A_\alpha$ in box/product topology, since each $A_\alpha$ is a subspace themselves. □

**Theorem 19.8.** *If each space $X_\alpha$ is Hausdorff then $\prod X_\alpha$ is Hausdorff in both the box and the product topologies.*

*Proof.* If $x, y \in \prod X_\alpha$ and $x \neq y$, then $\exists \beta \in J$ s.t. $x_\beta \neq y_\beta$. As a result, $\exists U, V$ neighborhoods of $x_\beta, y_\beta$ s.t. $U \cap V = \varnothing$ by the Hausdorff property.

Thus, $\prod U_\alpha$ and $\prod V_\alpha$ would be disjoint neighborhoods for $x$ and $y$, where $U_\alpha = V_\alpha = X_\alpha$ except for $\beta$, where $U_\beta = U$ and $V_\beta = V$. □

**Theorem 19.9.** *Let $X_\alpha$ be an indexed family of spaces. Let $A_\alpha \subset X_\alpha$ for each $\alpha$. If $\prod X_\alpha$ is given either the product or the box topology, then*

$$\prod \bar{A}_\alpha = \overline{\prod A_\alpha}.$$

*Proof.* Suppose $\mathbf{x} \in \prod \bar{A}_\alpha$. Notice that for any basis element $\prod U_\alpha$ that contains $\mathbf{x}$, $\exists y_\alpha \in U_\alpha \cap A_\alpha$. Therefore, $\mathbf{y} = \{y_\alpha\} \in \prod U_\alpha \cap \prod A_\alpha$, so $\mathbf{x} \in \overline{\prod A_\alpha}$.

On the other hand, suppose $\mathbf{x} \in \overline{\prod A_\alpha}$, then we want to show that for any $\beta \in J$ and $U_\beta$ neighborhood of $x_\beta$, $x_\beta \in \bar{A}_\beta$.

Notice that $V = \pi_\beta^{-1}(V_\beta)$ is a neighborhood of $\mathbf{x}$, so $V \cap \prod A_\alpha \neq \varnothing$. Suppose $\mathbf{y}$ is in the intersection, then $y_\beta \in V_\beta \cap A_\beta$. If $x_\beta = y_\beta$, then $x_\beta \in A_\beta \subset \bar{A}_\beta$. Otherwise, $x_\beta \in \bar{A}_\beta$ as it is a limit point. Therefore, $\mathbf{x} \in \prod \bar{A}_\alpha$. □

**Theorem 19.10.** *Let $f : A \to \prod_{\alpha \in J} X_\alpha$ be given by the equation*

$$f(a) = (f_\alpha(a))_{\alpha \in J}$$

*where $f_\alpha : A \to X_\alpha$ for each $\alpha$. Let $\prod X_\alpha$ have the product topology. Then the function $f$ is continuous iff $f_\alpha$ is continuous for all $\alpha$.*

*Proof.* Let $\pi_\beta$ be the projection of the product onto its $\beta$-th factor. The function $\pi_\beta$ is continuous, for if $U_\beta$ is open in $X_\beta$ then $\pi_\beta^{-1}(U_\beta) = \prod U_\alpha$ where $U_\alpha = X_\alpha$ for all $\alpha$ except $\beta$. Thus, $\pi_\beta^{-1}(U_\beta)$ is open in $\prod X_\alpha$. Thus, $f_\beta = \pi_\beta \circ f$, being the composite of two continuous functions, is continuous.

On the other hand, suppose each $f_\alpha$ is continuous. Let $U = \prod U_\alpha$ be a basis element in $\prod X_\alpha$. Notice that $f^{-1}(U) = \bigcap f_\alpha^{-1}(U_\alpha)$ is open, as $f_\alpha^{-1}(U_\alpha) \neq A$ for finitely many $\alpha$, so $f^{-1}(U)$ is an intersection of finitely many open sets and is open. □

**Remark 19.11.** *The second part of the proof does not work for the box topology, since clearly the inverse of $f^{-1}(U)$ might be an infinite intersection of open sets and is not necessarily open.*

*A counterexample for when the theorem fails for the box topology is to consider $f : \mathbb{R} \to \mathbb{R}^\omega$ where*

$$f(t) = (t, t, ...).$$

*Each of $f_n : \mathbb{R} \to \mathbb{R}$ is continuous, but $f$ is not, since a basis element for the box topology,*

$$(-1, 1) \times (-\tfrac{1}{2}, \tfrac{1}{2}) \times ...$$

*has the inverse image $0$, which is not open in $\mathbb{R}$.*

## Exercises

**Problem 6.** *Let $\mathbf{x_1}$, $\mathbf{x_2}$,... be a sequence of points in $\prod X_\alpha$. Show that the sequence converges to the point $\mathbf{x}$ iff the sequence $\pi_\alpha(\mathbf{x_1})$, $\pi_\alpha(\mathbf{x_2})$,... converges to $\pi_\alpha(\mathbf{x})$ for each $\alpha$. Is it true for the box topology on $\prod X_\alpha$?*

*Proof.* (i) Suppose that $\mathbf{x_1}$, $\mathbf{x_2}$,... converge to $\mathbf{x}$. Let $\beta$ be fixed.

For any neighborhood of $\pi_\beta(\mathbf{x})$, namely $U_\beta$, $\prod U_\alpha$ is a neighborhood of $\mathbf{x}$ in the product topology if $U_\alpha = X_\alpha$ for all $\alpha$ except $\beta$.

Thus, $\exists N \in \mathbb{N}$ s.t. $n > N \Rightarrow \pi_\beta(\mathbf{x_n}) \in U_\beta$, which makes $\pi_\beta(\mathbf{x_1})$, $\pi_\beta(\mathbf{x_2})$,... converging to $\pi_\beta(\mathbf{x})$.

(ii) On the other hand, suppose that $\pi_\alpha(\mathbf{x_1})$, $\pi_\alpha(\mathbf{x_2})$,... converge to $\pi_\alpha(\mathbf{x})$ for each $\alpha$. Then for any basis element $U = \prod U_\alpha$ of $\mathbf{x}$, let $\{n_1, ..., n_k\}$ be the set s.t. $U_{\alpha_{n_i}} \neq X_{\alpha_{n_i}}$.

Therefore, $\exists N_1, ..., N_k$ s.t. $n \geq N_i \Rightarrow \pi_{\alpha_i}(\mathbf{x_n}) \in U_{\alpha_{n_i}}$.

Let $N$ be the maximum of $N_i$'s, then $n \geq N \Rightarrow \mathbf{x_n} \in \prod U_\alpha = U$, indicating that the sequence $\mathbf{x_1}$, $\mathbf{x_2}$,... converges to the point $\mathbf{x}$.

(iii) The if direction is not correct for the box topology. Let $\mathbf{x_n} = (-\frac{1}{n}, -\frac{1}{n}, ...)$, $\mathbf{x} = (0, 0, ...)$ then $U =$

$$(-1, 1) \times (-\tfrac{1}{2}, \tfrac{1}{2}) \times ...$$

is a basis neighborhood of $\mathbf{x}$ and the sequence $\mathbf{x_n}$ does not converge to $\mathbf{x}$ even though each projection converges (since $\nexists n \in \mathbb{N}$ s.t. $\mathbf{x_n} \in U$).

□

**Problem 7.** *Let $\mathbb{R}^\infty$ be subset of $\mathbb{R}^\omega$ consisting of all sequences s.t. $x_i \neq 0$ for only finitely many values of $i$. What is the closure of $\mathbb{R}^\infty$ in the box and the product topology of $\mathbb{R}^\omega$?*

*Proof.* (i) In the box topology of $\mathbb{R}^\omega$, $\overline{\mathbb{R}^\infty} = \mathbb{R}^\infty$. For any $\mathbf{x} \notin \mathbb{R}^\infty$, there are infinitely many $i$ s.t. $x_i \neq 0$. Thus, $\exists U_i$ open in $\mathbb{R}_i$ s.t. $x_i \in U_i$, $0 \notin U_i$. Thus, take $U = \prod U_i$ where $U_i = \mathbb{R}$ if $x_i = 0$, then $U \cap \mathbb{R}^\infty = \varnothing$ since any element in $U$ has infinitely many indices not equal to 0.

(ii) In the product topology, $\overline{\mathbb{R}^\infty} = \mathbb{R}^\omega$. For any neighborhood $U$ of $\mathbf{x} \notin \mathbb{R}^\infty$, $\exists$ only finitely many $U_i$ that do not equal to $\mathbb{R}$. Thus, $\mathbf{y} = (y_n)$ with $y_i = x_i$ if $U_i \neq \mathbb{R}$ and $y_i = 0$ for other $i$ is in $U \cap \mathbb{R}^\infty$, so $\overline{\mathbb{R}^\infty} = \mathbb{R}^\omega$.

□

## 20 The Metric Topology

**Definition 20.1.** *A **metric** on s set $X$ is a function*

$$d : X \times X \longrightarrow \mathbb{R}$$

*with the following properties:*

*(i) $d(x, y) \geq 0$ for all $x, y \in X$; equality holds iff $x = y$.*

*(ii) $d(x, y) = d(y, x)$ for all $x, y \in X$.*

*(iii) $d(x, y) + d(y, z) \geq d(x, z), \forall x, y, z \in X$.*

*$d(x, y)$ is called the **distance** between $x$ and $y$ in the metric $d$.*

*Given $\epsilon > 0$, the set*

$$B_d(x, \epsilon) = \{y \mid d(x, y) < \epsilon\}$$

*is called the $\epsilon$**-ball centered at $x$**.*

**Definition 20.2.** *Let $d$ be a metric on the set $X$. The collection of all $\epsilon$-balls $B_d(x, \epsilon)$, for $x \in X$ and $\epsilon > 0$, forms a basis for a topology on $X$, called the **metric topology** on $X$ induced by $d$.*

To check that the $\epsilon$-balls indeed form a basis on $X$, we first show that if $y \in B(x, \epsilon)$, then $\exists B(y, \delta)$ centered at $y$ s.t. $B(y, \delta) \subset B(x, \epsilon)$ by taking $\delta = \epsilon - d(x, y)$.

Notice that the first condition for basis is trivial here. Now to check the second condition for a basis, let $B_1$ and $B_2$ be two basis element and $y \in B_1 \cap B_2$. Then, $\exists \delta_1$, $\delta_2$ s.t. $B(y, \delta_1) \subset B_1$, $B(y, \delta_2) \subset B_2$. Taking $\delta = \min(\delta_1, \delta_2)$, we have $B(y, \delta) \subset B_1 \cap B_2$.

**Definition 20.3.** *A topological space $X$ is called **metrizable** if there exists a metric $d$ on the set $X$ that induces the topology of $X$. A **metric space** is a metrizable space $X$ together with a specific metric $d$ that gives the topology of $X$.*

**Definition 20.4.** *Let $X$ be a metric space with metric $d$. A subset $A$ of $X$ is said to be **bounded** if $\exists M$ s.t.*

$$d(a_1, a_2) \leq M$$

*for any $a_1$, $a_2 \in X$.*

*If $X$ is bounded and non-empty, the **diameter** of $A$ is defined to be the number*

$$\text{diam } A = \sup\{d(a_1, a_2) \mid a_1, a_2 \in A\}.$$

**Theorem 20.5.** *Let $X$ be a metric space with metric $d$. Define*

$$\bar{d} = \min\{d(x, y), 1\}.$$

*Then $\bar{d}$ is a metric that induces the same topology as $d$.*

*The metric $\bar{d}$ is called the **standard bounded metric** corresponding to $d$.*

It is not difficult to check that $\bar{d}$ is a metric. The third condition (triangle inequality) can be proven by separating the case of whether $d(x, y) \geq 1$ or not (and $d(y, z)$).

To prove that the metric topology induced by $\bar{d}$ is the same as the one induced by $d$, we use the fact that the two metrics have the same collection of balls $B(x, \epsilon)$ with $\epsilon < 1$, which would generate the same topology as the set of all balls.

**Definition 20.6.** *Given $\mathbf{x} = (x_1, ..., x_n) \in \mathbb{R}^n$, we define the **norm** of $\mathbf{x}$ by the equation*

$$||x|| = (x_1^2 + ... + x_n^2)^{\frac{1}{2}}$$

*and the **euclidean metric** $d$ on $\mathbb{R}^n$ is defined by the equation*

$$d(\mathbf{x}, \mathbf{y}) = ||\mathbf{x} - \mathbf{y}|| = [(x_1 - y_1)^2 + ... + (x_n - y_n)^2]^{\frac{1}{2}}$$

*The **square metric** is defined as*

$$\rho(\mathbf{x},\mathbf{y}) = max\{|x_1 - y_1|, ..., |x_n - y_n|\}.$$

Showing $d$ is a metric requires some work with inequality, and the process is shown in the exercises, while showing $\rho$ is a metric is quite trivial.

**Lemma 20.7.** *Let $d$ and $d'$ be two metrics on the set $X$; let $\mathcal{T}$ and $\mathcal{T}'$ be the topologies they induce. Then $\mathcal{T}'$ is finer than $\mathcal{T}$ iff for any $x \in X$, $\epsilon > 0$, $\exists \delta > 0$ s.t.*

$$B_{d'}(x,\delta) \subset B_d(x,\epsilon)$$

The above lemma can easily be proved by using **Lemma 13.5**.

**Theorem 20.8.** *The topologies on $\mathbb{R}^n$ induced by the euclidean metric $d$ and the square metric $\rho$ are the same as the product topology on $\mathbb{R}^n$.*

*Proof.* Let $\mathbf{x} = (x_1, ..., x_n)$ and $\mathbf{y} = (y_1, ..., y_n)$. Notice that we have

$$\rho(\mathbf{x},\mathbf{y}) \leq d(\mathbf{x},\mathbf{y}) \leq \sqrt{n}\rho(\mathbf{x},\mathbf{y}),$$

which shows us that

$$B_d(\mathbf{x},\epsilon) \subset B_\rho(\mathbf{x},\epsilon),$$
$$B_\rho(\mathbf{x},\tfrac{\epsilon}{\sqrt{n}}) \subset B_d(\mathbf{x},\epsilon).$$

Thus, the two topologies are the same following the preceding lemma.

Now we show that the product topology on $\mathbb{R}^n$ is the same as the topology induced by $\rho$. Take a basis element for the product topology on $\mathbb{R}^n$

$$B = (a_1, b_1) \times ... \times (a_n, b_n).$$

Let $\mathbf{x} = (x_1, ..., x_n)$ be contained in $B$.

For any $i$, $\exists \epsilon_i$ s.t. $(x_i - \epsilon_i, x_i + \epsilon_i) \subset (a_i, b_i)$. Thus, $B_\rho(\mathbf{x},\epsilon) \subset B$ where $\epsilon$ is the minimum among $\epsilon_1, ..., \epsilon_n$. Thus, the metric topology is finer than the product topology.

On the other hand, a basis element for the metric topology, $B_\rho(x,\epsilon) = (x_1 - \epsilon, x_1 + \epsilon) \times ... \times (x_n - \epsilon, x_n + \epsilon)$, is itself a basis element for the product topology. Thus, the product topology is the same as the metric topology generated by $\rho$. □

Notice that if we directly try to generalize $d$ and $\rho$ on $\mathbb{R}^\omega$, the expression we gain do not always make sense. To make sure that the metric always has a value, we define

**Definition 20.9.** *Given an index set $J$ and $\mathbf{x} = (x_\alpha)_{\alpha \in J}$ and $\mathbf{y} = (y_\alpha)_{\alpha \in J}$, we define a metric $\bar{\rho}$ on $\mathbb{R}^J$ by the equation*

$$\bar{\rho}(\mathbf{x},\mathbf{y}) = \sup\{\bar{d}(x_\alpha, y_\alpha) \mid \alpha \in J\}.$$

*where $\bar{d}$ is defined in* **Theorem 20.5**. *The definition ensures that the value $\bar{\rho}(\mathbf{x},\mathbf{y})$ is bounded. Then, $\bar{\rho}$ is a metric with the name **uniform metric** on $\mathbb{R}^J$, the topology it induces is called the **uniform topology**.*

**Theorem 20.10.** *The uniform topology on $\mathbb{R}^J$ is finer than the product topology and coarser than the box topology; if $J$ is infinite, all three topologies are different.*

*Proof.* Suppose $\mathbf{x} = (x_\alpha)_{\alpha \in J}$ is in a product basis element $\prod U_\alpha$. Let $\alpha_1, ..., \alpha_n$ be the indices where $U_{\alpha_i} \neq \mathbb{R}$. For each $i$, choose $\epsilon_i$ s.t. $B(x_{\alpha_i}, \epsilon_i) \subset U_{\alpha_i}$. Then let $\epsilon = \min\{\epsilon_1, ..., \epsilon_n\}$, then $B_{\bar{\rho}}(\mathbf{x},\epsilon) \subset \prod U_\alpha$ and contains $\mathbf{x}$. Thus, the uniform topology is finer than the product topology.

On the other hand, for any $\epsilon$-ball centered at $\mathbf{x}$, the product

$$U = \prod(x_\alpha - \frac{1}{2}\epsilon, x_\alpha + \frac{1}{2}\epsilon)$$

is a basis element for the box topology containing $\mathbf{x}$ that is contained in the $\epsilon$-ball. Thus, the box topology is finer than the uniform topology. □

It turns out that when $J$ is infinite, the only case where $\mathbb{R}^J$ is metrizable is where $J$ is countable and $\mathbb{R}^J$ has the product topology.

**Theorem 20.11.** *Let $\bar{d}(a,b) = \min(|a-b|, 1)$ be the standard bounded metric on $\mathbb{R}$. If $\mathbf{x}$ and $\mathbf{y}$ are two points of $\mathbb{R}^\omega$, define*

$$D(\mathbf{x},\mathbf{y}) = \sup\{\frac{\bar{d}(x_i,y_i)}{i}\}.$$

*Then $D$ is a metric that induces the product topology on $\mathbb{R}^\omega$.*

*Proof.* (i) We first show that the triangle inequality holds for $D$. Notice that for all $i$

$$\frac{\bar{d}(x_i,z_i)}{i} \le \frac{\min(d(x_i,y_i)+d(y_i,z_i),1)}{i} \le \frac{\bar{d}(x_i,y_i)}{i} + \frac{\bar{d}(y_i,z_i)}{i} \le D(\mathbf{x},\mathbf{y}) + D(\mathbf{y},\mathbf{z}).$$

Thus,

$$\sup\{\frac{\bar{d}(x_i,y_i)}{i}\} \le D(\mathbf{x},\mathbf{y}) + D(\mathbf{y},\mathbf{z}).$$

Thus, $D$ is a metric.

(ii) To show that $D$ induces the product topology, we first consider a basis in the metric topology that contains $\mathbf{x}$ and is contained in an open set $U$. Let the basis be $B_D(\mathbf{x},\epsilon)$. Now, notice that there is some $n \in \mathbb{N}$ s.t. $\frac{1}{n} \le \epsilon$. Let $V$ be the basis $(x_1-\epsilon, x_1+\epsilon) \times ... \times (x_n - \epsilon, x_n+\epsilon) \times \mathbb{R} \times ...$, then we have $x \in V \subset B_D(\mathbf{x},\epsilon) \subset U$, so the product topology is finer than the metric topology.

(iii) Consider the basis $U = \prod U_i$ in the product topology, with $U_i \neq \mathbb{R}$ at indices $\alpha_1, ..., \alpha_n$. Let $\mathbf{x} \in U$.
Notice that there is some $\epsilon_i$ s.t. $(x-\epsilon_i, x+\epsilon_i) \subset U_{\alpha_i}$. Let $\epsilon$ be the minimum among the values $\epsilon_i/i$. We then have $\mathbf{x} \in B_D(\mathbf{x},\epsilon) \subset U$.

Therefore, the two topologies are the same. □

## Exercises

**Problem 2.** *Show that $\mathbb{R}\times\mathbb{R}$ in the dictionary order topology is metrizable.*

*Proof.* Notice that the dictionary order topology is the same as the topology on $\mathbb{R}_d \times \mathbb{R}$, where $\mathbb{R}_d$ has the discrete topology. Also notice that $\mathbb{R}_d$ is metrizable by the discrete metric.

Thus, it suffices to show that the product of two metrizable spaces is still metrizable. Suppose we have $d_1$, $d_2$ the metrics on the space $X$ and $Y$, then $\rho(x,y) = \max(d_1(x,y), d_2(x,y))$ is a new metric.

To show $\rho$ induces the product topology, notice that we have

$$B_\rho(x,\epsilon) \subset B_{d_1}(x_1,\epsilon_1) \times B_{d_2}(x_2,\epsilon_2) \subset B_\rho(x,\epsilon')$$

on the basis of the topologies, where $\epsilon = \min(\epsilon_1,\epsilon_2)$ and $\epsilon' = \max(\epsilon_1,\epsilon_2)$. □

**Problem 3.** *Show that $d: X \times X \to \mathbb{R}$ is continuous if $X$ is a metric space with metric $d$.*

*Proof.* Consider a basis element $(a,b) \subset \mathbb{R}$. Let $U = d^{-1}((a,b))$ and let $(x,y) \in U$.

Let $\epsilon$ be the minimal of $(d(x,y)-a)/2$ and $(b-d(x,y))/2$. We then have

$$B_d(x,\epsilon) \times B_d(y,\epsilon) \subset U,$$

implying that $U$ is open. □

**Problem 4.** *Consider the product, uniform, and box topologies on $\mathbb{R}^\omega$.*

*In which topologies are the following functions from $\mathbb{R}$ to $\mathbb{R}^\omega$ continuous?*

$$f(t) = (t, 2t, 3t, ...),$$
$$g(t) = (t, t, t, ...),$$
$$h(t) = (t, \tfrac{t}{2}, \tfrac{t}{3}, ...).$$

*Proof.* All three functions are continuous in the product topologies since all of their components are.

All three functions are not continuous in the box topology since the inverse image of

$$(-1, 1) \times (-\tfrac{1}{2^2}, \tfrac{1}{2^2}) \times (-\tfrac{1}{3^3}, \tfrac{1}{3^3}) \times ...$$

is always $\{0\}$ which is not open in $\mathbb{R}$.

$f(t)$ is not continuous in the uniform topology, since the inverse image of $B_{\bar{\rho}}(\mathbf{0}, 1)$ is $\{0\}$, which is not open in $\mathbb{R}$. However, $g(t)$ and $h(t)$ are both continuous since the inverse image of any neighborhood $B_{\bar{\rho}}(g(t), \epsilon)$ or $B_{\bar{\rho}}(h(t), \epsilon)$, is $(t - \epsilon, t + \epsilon)$ which is open in $\mathbb{R}$. ☐

**Problem 5.** *What is the closure of $\mathbb{R}^\infty$ in $\mathbb{R}^\omega$ in the uniform topology?*

*Proof.* Let $\mathbf{x} \in \overline{R^\infty}$. Then by definition, if $\mathbf{x} \notin \mathbb{R}^\infty$, then for any $\epsilon > 0$, $B_{\bar{\rho}}(\mathbf{x}, \epsilon) \cap \mathbb{R}^\infty \neq \varnothing$.

Notice that we only need to consider $\epsilon$ that is sufficiently small, and when $\epsilon < 1$, $\bar{\rho}$ and $\rho$ makes no difference, if we define $\rho(\mathbf{x}, \mathbf{y})$ to be $\sup\{|x_i - y_i|\}$. (This is not a very accurate definition as the metric is not well-defined sometimes; but in the neighborhood it is).

Therefore, we have that $B_\rho(\mathbf{x}, \epsilon) \cap \mathbb{R}^\infty \neq \varnothing$ for any $\epsilon > 0$.

In other words, we have that for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. for any $n \geq N$, $0 \in (x_n - \epsilon, x_n + \epsilon)$.

Now, notice that this is exactly the same as the definition for $\{x_n\}$ to converge to 0.

Therefore, $\overline{\mathbb{R}^\infty}$ under the uniform topology is the same as the set of all sequences converging to 0.

Notice that this includes but is not exactly the set $\mathbb{R}^\infty$. For instance, $(1, \frac{1}{2}, ...)$ is not in $\mathbb{R}^\infty$ but is in $\overline{\mathbb{R}^\infty}$. ☐

**Problem 6.** *Let $\bar{\rho}$ be the uniform metric on $\mathbb{R}^\omega$. Giving $\mathbf{x} = (x_1, x_2, ...) \in \mathbb{R}^\omega$ and given $0 < \epsilon < 1$, let*

$$U(\mathbf{x}, \epsilon) = (x_1 - \epsilon, x_1 + \epsilon) \times ...$$

*(i) Show that $U(\mathbf{x}, \epsilon)$ is not equal to $B_{\bar{\rho}}$.*

*(ii) Show that $U(\mathbf{x}, \epsilon)$ is not open in the uniform topology.*

*(iii) Show that*

$$B_{\bar{\rho}}(\mathbf{x}, \epsilon) = \bigcup_{\delta < \epsilon} U(\mathbf{x}, \delta).$$

*Proof.* (i) Notice that if $\bar{\rho}(\mathbf{x}, \mathbf{y}) < \epsilon$, then $|x_i - y_i| \leq \epsilon$ for any $i$. Therefore, $B_{\bar{\rho}}(\mathbf{x}, \epsilon) \subset U(\mathbf{x}, \epsilon)$.

On the other hand, the converse is not always true. For instance, consider the neighborhood $U(0, \frac{1}{2})$. $(\frac{1}{2} - \frac{1}{2}) \times (\frac{1}{2} - \frac{1}{3}) \times ...$ is in $U(0, \frac{1}{2})$ but is not in $B_{\bar{\rho}}(0, \frac{1}{2})$ as $\sup\{|x_i - y_i|\} = \frac{1}{2}$.

Therefore, $B_{\bar{\rho}}(\mathbf{x}, \epsilon) \subsetneq U(\mathbf{x}, \epsilon)$.

(ii) We directly apply the very first definition of open sets.

Notice that in the above example, the element $(\frac{1}{2} - \frac{1}{2}) \times (\frac{1}{2} - \frac{1}{3}) \times ...$ is not in any basis element $B$ for the uniform topology that is contained in $U(0, \frac{1}{2})$.

(iii) RHS$\subset$ LHS is quite trivial.

To prove the reverse, suppose $\mathbf{y} \in B_{\bar{\rho}}(\mathbf{x}, \epsilon)$. Let $\delta = \sup\{|x_i - y_i|\}$, then $\delta < \epsilon$ and $\mathbf{y} \in U(\mathbf{x}, \delta)$, as $|x_i - y_i| \leq \delta$.

☐

## 21 The Metric Topology (Cont'd)

**Theorem 21.1.** *Let $f : X \to Y$; let $X$ and $Y$ be metrizable with metrics $d_X$ and $d_Y$. Then the continuity of $f$ is equivalent to the requirement that for any $\epsilon > 0$, $\exists\delta > 0$, s.t.*

$$d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \epsilon.$$

*Proof.* (i) Suppose that $f$ is continuous. For any $\epsilon > 0$, $B_{d_Y}(f(x), \epsilon)$ is a neighborhood of $f(x)$ and hence has an open inverse image $f^{-1}(B_{d_Y}(f(x), \epsilon))$. Therefore, there is a basis element $B_{d_X}(x, \delta) \in f^{-1}(B_{d_Y}(f(x), \epsilon))$.

In other words, if $y \in B_{d_X}(x, \delta)$, then $d_Y(f(x), f(y)) < \epsilon$, as we desired.

(ii) Suppose that the condition is met. Let $V$ be an open set and $x \in f^{-1}(V)$. We then know that $f(x) \in V$ and, therefore, $\exists\epsilon > 0$ s.t. $B_{d_Y}(f(x), \epsilon) \subset V$.

By the condition, we have some $\delta > 0$ s.t. $f(B_{d_X}(x, \delta)) \subset B_{d_Y}(f(x), \epsilon)$. Hence, $B_{d_X}(x, \delta) \subset f^{-1}(V)$ and contains $x$. Therefore, $f^{-1}(V)$ is open.

□

**Lemma 21.2.** *Let $X$ be a topological space, $A \subset X$. If there is a sequence of points of $A$ converging to $x$, then $x \in \bar{A}$. The converse holds if $X$ is metrizable.*

*Proof.* (i) The first direction is trivial, as any neighborhood of $x$ contains some points of $A$ (or $x \in A$ itself).

(ii) Conversely, suppose that $X$ is metrizable with metric $d$. Then if $x \in A$, $x_n = x$ for any $n$ is clearly a sequence converging to $x$. If $x \notin A$, then there exists $x_n \in B_d(x, \frac{1}{n})$, and $x_n$ would be a sequence converging to $x$, as any basis element $B_d(x, \epsilon)$ contains $x_n$ for any $n \geq N$, where $\epsilon > \frac{1}{N}$.

□

**Theorem 21.3.** *Let $f : X \to Y$. If the function $f$ is continuous, then for every sequence $x_n$ converging to $x$, we have $f(x_n)$ converging to $f(x)$. The converse is true if $X$ is metrizable.*

*Proof.* (i) Suppose that $x_n$ converges to $x$. Then for any neighborhood $U$ containing $f(x)$, $f^{-1}(U)$ is open and therefore contains $x_n$ for any $n \geq N$ where $N$ is some integer. Therefore, $f(x_n) \in V$ and $f(x_n)$ therefore converges to $f(x)$.

(ii) Suppose that $X$ is metrizable with metric $d$. We want to show that $f(\bar{A}) \subset \overline{f(A)}$, which is equivalent to $f$ being continuous.

Suppose that $x$ is a limit point of $A \subset X$, then for there exists a sequence $x_n$ converging to $x$ by the above lemma.

Therefore, we know that $f(x_n)$ converges to $f(x)$ by the condition and therefore $f(x) \in \overline{f(A)}$ according to the the above lemma. Therefore, $f(\bar{A}) \subset \overline{f(A)}$ and $f$ is continuous.

□

**Definition 21.4.** *Notice that in the proofs above, we did not use the full strength of metric spaces. Instead, we only need a set of neighborhoods $\{U_n\}$ of $x$ s.t. any neighborhood $U$ of $x$ contains some $U_n$.*

*If such $\{U_n\}$ exists at $x$, then $X$ is called having a* ***countable basis at x****. A space that has a countable basis at each $x \in X$ is said to satisfy the* ***first countability axiom****.*

**Lemma 21.5.** *The addition, subtraction, and multiplication operations are continuous functions from $\mathbb{R} \times \mathbb{R}$ to $\mathbb{R}$. The quotient operation is continuous from $\mathbb{R} \times (\mathbb{R} - \{0\})$ to $\mathbb{R}$.*

The lemma can be proved by using the $\epsilon - \delta$ definition of continuity given in **Theorem 21.1**.

**Theorem 21.6.** *Let $X$ be a topological space. If $f, g : X \to \mathbb{R}$ are continuous functions, then $f+g$, $f-g$, $fg$ are continuous. If $g(x) \neq 0$ for all $x$, then $\frac{f}{g}$ is also continuous.*

This theorem can be proved by considering the operations on $f$ and $g$ as a composite of continuous functions, $h : X \to \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, and a projection $h(x) = f(x) \times g(x)$.

**Definition 21.7.** *Let $f_n : X \to Y$ be a sequence of functions from $X$ to the metric space $Y$. Let $d$ be the metric of $Y$. $(f_n)$ is said to* ***converge uniformly*** *to the function $f : X \to Y$ if given $\epsilon > 0$, $\exists N \in \mathbb{N}$ s.t.*

$$d(f_n(x), f(x)) < \epsilon$$

*for all $n > N$ and $x \in X$.*

**Theorem 21.8.** *Let $f_n : X \to Y$ be a sequence of continuous functions from the topological space $X$ to the metric space $Y$. If $(f_n)$ converges uniformly to $f$, then $f$ is continuous.*

*Proof.* Let $V$ be an open set in $Y$ and $x_0 \in f^{-1}(V)$. We want to show that there exists a neighborhood $U$ containing $x_0$ s.t. $f(U) \subset V$.

Let $B(f(x_0), \epsilon) \subset V$, we want to show that there exists $\delta > 0$ s.t. $f(x) \in B(f(x_0), \epsilon)$ if $d(x, x_0) < \delta$.

Notice that since each $f_n$ is continuous, there exists some $\delta > 0$ s.t. if $d(x_0, x) < \delta$ then $d(f_n(x_0), f_n(x)) < \frac{\epsilon}{3}$ for some fixed $n$.

Additionally as $(f_n)$ converges uniformly to $f$, there exists some $N \in \mathbb{N}$ s.t. if $n \geq N$, then $d(f(x), f_n(x)) < \frac{\epsilon}{3}$ and $d(f(x_0), f_n(x_0)) < \frac{\epsilon}{3}$.

We therefore have that $d(f(x), f(x_0)) < \epsilon$ by triangle inequality (and fix some $n$) if $d(x, x_0) < \delta$. In other word, $f(B(x_0, \delta)) \subset B(f(x_0), \epsilon)$ and $f$ is therefore continuous. □

**Remark 21.9.** *Let $\mathbb{R}^X$ be the set of all functions $f : X \to \mathbb{R}$, it has a metric $\bar{\rho}$ defined by $\bar{\rho}(f, g) = \sup\{\bar{d}(f_x, g_x)\} = \sup\{\bar{d}(f(x), g(x))\}$.*

*Therefore, $(f_n)$ converges to $f$ uniformly iff it converges to $f$ in the metric space $(\mathbb{R}^X, \rho)$.*

**Corollary 21.10.** *$\mathbb{R}^\omega$ in the box topology is not metrizable.*

*Proof.* We shall show that the sequence lemma fails. In other words, find some $x \in \bar{A}$ but there is no sequence converging to $x$.

Consider $A \subset \mathbb{R}^\omega$ where $A = \{(x_1, x_2, ...) \mid x_i > 0\}$. Clearly $\mathbf{0}$ is a limit point of $A$.

However, there does not exist a sequence $(a_n)$ converging to $x$, since we can let $B = (-a_{1,1}, a_{1,1}) \times (-a_{2,2}, a_{2,2}) \times ...$ is a neighborhood of $\mathbf{0}$ that does not contain any point $a_n$, if a term $a_n$ of the sequence is denoted as $(a_{n,1}, a_{n,2}, ...)$. □

**Theorem 21.11.** *An uncountable product of $\mathbb{R}$ with itself is not metrizable.*

*Proof.* Let $A$ be the subset of $\mathbb{R}^J$ consisting all the points of $\mathbb{R}^J$ where all indices equals 1 except for finitely many indices.

$\mathbf{0}$ belongs to the closure of $A$ as any basis element $\prod U_\alpha$ containing $\mathbf{0}$ has $U_\alpha = \mathbb{R}$ except for finitely many $\alpha$. Thus, there are always elements of $A$ intersecting it, namely $\mathbf{x}$ where $x_\alpha = 1$ iff $U_\alpha = \mathbb{R}$, and 0 otherwise.par On the other hand, for any sequence $(a_n)$ in $A$, let $J_n$ denote the set of all indices $\alpha$ for which the $\alpha$-th index of $a_n$ is different from 1. $\bigcup\limits_{n \in \mathbb{N}} J_n$ is a countable union of finite sets so it is also countable. As $J$ is uncountable, $\exists \beta \in J$ that does not lie in any of $J_n$. Then,

$$U = \prod U_\alpha \text{ where } U_\alpha = \mathbb{R} \text{ except for } U_\beta \text{ which equals } (-1, 1)$$

is a basis element containing $\mathbf{0}$ that does not intersect any $a_n$. Thus, no sequence converges to $\mathbf{0}$ and $\mathbb{R}^J$ is uncountable. □

## Exercises

**Problem 3.** *Let $(X_n)$ be a finite sequence of spaces with metric $d_n$ on each $X_n$. Show that*

$$\rho(x, y) = \max\{d_i(x_i - y_i)\}$$

*is a metric that induces the product topology on $\prod X_n$.*

*Proof.* Apparently $\rho$ is a metric.

Consider an arbitrary basis element of the product topology on $\prod X_n$ $B = (x_1 - \epsilon_1, x_1 + \epsilon_1) \times ... \times (x_n - \epsilon_n, x_n + \epsilon_n)$.

Let $\epsilon$ be the minimum of $\epsilon_1, ..., \epsilon_n$. Then $B_\rho(\mathbf{x}, \epsilon)$ is a subset of $B$ open in the metric topology, so the metric topology is finer than the product topology.

On the other hand, for any $\epsilon > 0$, $B_\rho(\mathbf{x}, \epsilon) = (x_1 - \epsilon, x_1 + \epsilon) \times ... \times (x_n - \epsilon, x_n + \epsilon)$ is itself open in $\prod X_n$, so the metric topology is coarser the product topology.

Therefore, the two topologies are the same. □

**Problem 7.** *Let $X$ be a topological space and let $Y$ be a metric space. Let $f_n : X \to Y$ be a sequence of continuous functions. Let $(x_n)$ be a sequence of points converging to $x$. prove that if the sequence $(f_n)$ uniformly converges to $f$, then $(f_n(x_n))$ converges to $f(x)$.*

*Proof.* Suppose $(f_n)$ uniformly converges to $f$, then $f$ is continuous according to the theorem earlier.

Let $\epsilon > 0$, $U = B(f(x), \frac{\epsilon}{2})$ be an open set of $Y$ containing $f(x)$. $f$ continuous so $f^{-1}(U)$ is a neighborhood of $x$. As $(x_n)$ converges to $x$, $\exists N \in \mathbb{N}$ s.t. $n > N \Rightarrow x_n \in f^{-1}(U)$. As a result, $f(x_n) \in U$.

Since $(f(n))$ converges uniformly to $f$, $\exists N' \in \mathbb{N}$ s.t. $f_n(x_n) - f(x_n) < \frac{\epsilon}{2}$ if $n > N'$. Thus, $d(f_n(x_n), f(x)) < \epsilon$ so $f_n(x_n) \in U$ for all $n > \max(N, N')$. Thus, $(f_n(x_n))$ converges to $f(x)$. □

**Problem 11.** *Prove the following facts of* ***infinite series****.*

*(i) If $(s_n)$ is a bounded real sequence and $s_n < s_{n+1}$ for all $n \in \mathbb{N}$, then $(s_n)$ converges.*

*(ii) Let $(a_n)$ be a sequence of real numbers; define*

$$s_n = \sum_{i=1}^{\infty} a_i.$$

*If $s_n$ converges to $s$, then the infinite series $\sum_{i=1}^{\infty} a_i$ is said to converge to $s$. Then if $|a_i| < b_i$ and $\sum b_i$ converges, we have $\sum a_i$ converges.*

*(iii) Give a sequence of functions $(f_n) : X \to \mathbb{R}$, let*

$$s_n(x) = \sum_{i=1}^{n} f_i(x).$$

*Prove that if $|f_i(x)| \leq M_i$ for all $x \in X$ and $i$, and if $\sum M_i$ converges, we have $(s_n)$ converges uniformly to a function $s$.*

*Proof.* (i) Let $s$ be $\sup(s_n)$. Then for any $\epsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $n > N \Rightarrow s_n \in (s - \epsilon, s]$. Thus, $s_n$ converges to $s$.

(ii) $\sum |a_i|$ converges since the sequence of partial sums is bounded and non-decreasing. Similarly, $\sum |a_i| + a_i$ converges. As a result, their difference, $\sum a_i$, converges.

(iii) For any $x \in X$, the sum $s_n(x)$ converges according to the above part. Let the limit be $s(x)$.

To show that the sequence uniformly converges, take an arbitrary $\epsilon > 0$; as $\sum M_i$ converges, $\exists N \in \mathbb{N}$ s.t. $\sum_{N+1}^{\infty} M_i < \epsilon$. Thus, $\forall k > N$, $|s_k(x) - s_N(x)| = \sum_{i=N+1}^{k} |f_i(x)| < \sum_{N+1}^{\infty} M_i < \epsilon$.

Thus, $|s(x) - s_n(x)| < \epsilon$ for any $x \in X$ and $n \geq N$, so the sequence $(s_n)$ uniformly converges to $s$. □

## 22 The Quotient Topology

**Definition 22.1.** *Let $X$ and $Y$ be topological spaces; let $p : X \to Y$ be a surjective map. $p$ is called a* ***quotient map*** *if $U$ open in $Y$ is equivalent to $p^{-1}(U)$ open in $X$.*

**Definition 22.2.** *A subset $C$ of $X$ is* ***saturated*** *if $C$ is the complete inverse image of a subset of $Y$ (i.e., $C = p^{-1}(D)$ for some $D \subset Y$).*

*Thus, $p$ is a quotient map is equivalent to $p$ continuous and maps saturated open sets of $X$ to open sets of $Y$.*

**Definition 22.3.** *A map $f : X \to Y$ is an* ***open map*** *if for any $U \subset X$ open, $f(U)$ is open.*

***Closed map*** *is defined analogously.*

Any surjective continuous open or closed map is a quotient map, but the converse might not be true.

**Example 22.4.** *Let $\pi_1 : \mathbb{R}\times\mathbb{R} \to \mathbb{R}$ be the projection onto the first coordinate. $\pi_1$ is surjective and continuous and an open map since for any basis element $U \times V$ where $U$, $V$ are open in $\mathbb{R}$, then $\pi_1(U \times v) = U$ is open in $\mathbb{R}$. However, $\pi_1$ is not a closed map. Consider*

$$C = \{x \times y \mid xy = 1\}.$$

*$C$ is closed as it is the inverse image of the multiplication operation of a closed set $\{1\}$, but $\pi_1(C) = \mathbb{R}-\{0\}$ is not closed in $\mathbb{R}$.*

*Notice that if $A$ is a subspace of $\mathbb{R} \times \mathbb{R}$ with $A = C \cup \{0\}$, then $q : A \to \mathbb{R}$ by restricting $\pi_1$ on $A$ is continuous (as the restriction of a continuous function) and surjective, but it is not a quotient map.*

*$\{\mathbf{0}\}$ is open in $A$ as $\{\mathbf{0}\} = A \cap (-\frac{1}{2}, \frac{1}{2}) \times (-\frac{1}{2}, \frac{1}{2})$, and it is the only inverse of $\{0\}$, but $q(\mathbf{0}) = 0$ is not open in $\mathbb{R}$.*

**Definition 22.5.** *If $X$ is a space and $A$ is a set and $p : X \to A$ is surjective, then $\exists!$ topology $\mathcal{T}$ on $A$ relative to which $p$ is a quotient map, namely the set of subsets $U$ where $p^{-1}(U)$ is open in $X$. The resulted topology is called the* ***quotient topology*** *induced by $p$.*

It is not difficult to check that $\mathcal{T}$ is a topology.

**Definition 22.6.** *Let $X$ be a topological space, $X^*$ be a partition of $X$. Let $p : X \to X^*$ be the surjective map that carries each point of $X$ to the corresponding element in $X^*$ containing it. In the quotient topology induced by $p$, the space $X^*$ is called the* ***quotient space*** *of $X$.*

Notice that a subset $U$ of $X^*$ is open iff $p^{-1}(U)$ is open in $X$.

We can think of $X'$ by identifying some pair of equivalent points.

**Example 22.7.** *Let $D^2$ be the closed ball in $\mathbb{R}^2$:*

$$D^2 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\},$$

*and $S^1$ be its boundary $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$.*

*Then we can show that $D^2/S^1 \cong S^2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$.*

According to **Example 22.4**, the restriction of a quotient map on a subspace might not be a quotient map, but it must be a quotient map with some additional conditions.

**Theorem 22.8.** *Let $p : X \to Y$ be a continuous map; let $A$ be a subspace of $X$ that is saturated with respect to $p$; let $q : A \to p(A)$ be the restriction of $p$ on $A$. Then $q$ is a quotient map if:*

*(i) If $A$ is either open or closed in $X$.*

*(ii) If $p$ is either open or closed.*

*Proof.* Notice that $q$ is continuous as the restriction of a continuous function.

We start by verifying the following equations:

(i) $p^{-1}(V) = q^{-1}(V)$ if $V \subset p(A)$.
(ii) $p(U) \cap p(A) = p(U \cap A)$ for any $U \subset X$.

The first one is true because $A$ is saturated, so $p^{-1}(V)$ is contained in $A$ and therefore equals $q^{-1}(V)$.

Notice that clearly $p(U \cap A) \subset p(U) \cap p(A)$, and if $y \in p(U) \cap p(A)$, then there exists $x \in A$ and $u \in U$ s.t. $p(u) = p(x) = y$. Since $A$ is saturated, $u \in X$ so $u \in X \cap A$ so $p(U) \cap p(A) \subset p(U \cap A)$.

(i) Suppose that $A$ is open in $X$. Notice that $p(A)$ is open in $Y$. Therefore, for any $V \subset p(A)$ with $q^{-1}(V)$ open in $A$, $q^{-1}(V) = p^{-1}(V)$ is open in $X$ too, so $V$ is open in $Y$ and since it is a subset of $p(A)$, $V$ is open in $p(A)$.
(ii) Suppose that $p$ is open. For any $V \subset p(A)$ s.t. $q^{-1}(A)$ is open in $A$, we have that $q^{-1}(A) = A \cap U$ for some $U$ open in $X$. Therefore, $p(U)$ is open and $V = p(U \cap A) = p(U) \cap p(A)$ is open in $p(A)$.

Therefore, in either case $q$ is a quotient map. The proof for $A$ or $p$ closed is analogous. □

**Corollary 22.9.** *If $p : X \to Y$, $q : Y \to Z$ are quotient maps, then $q \circ p$ is also a quotient map.*

*Proof.* $q \circ p$ is continuous. For any $U \subset Z$ s.t. $(q \circ p)^{-1}(U)$ is open, $(q \circ p)^{-1}(U) = p^{-1}(q^{-1}(U))$ is open. Since $p$ and $q$ are quotient maps, $q^{-1}(U)$ and $U$ are open in $Y$ and $Z$, respectively. □

The product of a quotient map may not be a quotient map, and the quotient space of a space $X$ may not be Hausdorff even if $X$ is Hausdorff.

**Theorem 22.10.** *Let $p : X \to Y$ be a quotient map. Let $Z$ be a space and let $g : X \to Z$ be a map s.t. for any $y \in Y$, $g(p^{-1}(y))$ is a constant. Then $g$ induces a map $f : Y \to Z$ s.t. $f \circ p = g$.*

*The induced map $f$ is continuous iff $g$ is, and $f$ is a quotient map iff $g$ is one.*

$$\begin{tikzcd} X \arrow[d, "p"] \arrow[dr, "g"] & \\ Y \arrow[r, dashed, "f"] & Z \end{tikzcd}$$

*Proof.* Let $g(p^{-1}(y)) = z$, then if we set $f(y) = z$, we get a map $f : Y \to Z$ with $f \circ p = g$.

If $f$ is continuous or a quotient map, then $g$ is continuous or a quotient map as it is the composite of two continuous functions or quotient maps.

On the other hand, suppose $g$ is continuous, then $g^{-1}(U)$ is open in $X$ for any $U$ open in $Z$. Since $p$ is a quotient map, $p(g^{-1}(U)) = f^{-1}(U)$ is open in $Y$, so $f$ is continuous.

If $g$ is a quotient map, then $g$ is surjective so $f$ is surjective. Suppose that $V \subset Z$ s.t. $f^{-1}(V)$ is open in $Y$. Then $g^{-1}(V) = (pf)^{-1}(V)$ is open in $X$, so $V$ is open in $Z$, indicating that $f$ is a quotient map. □

**Theorem 22.11.** *Let $g : X \to Z$ be a surjective continuous map. Let $X^*$ be the following collection of subsets of $X$*

$$X^* = \{g^{-1}(\{z\}) \mid z \in Z\}.$$

*Give $X^*$ the quotient topology.*

*(i) The map $g$ induces a bijective continuous map $f : X^* \to Z$, which is a homeomorphism iff $g$ is a quotient map.*
*(ii) If $Z$ is Hausdorff, so is $X^*$.*

$$\begin{tikzcd} X \arrow[d, "p"] \arrow[dr, "g"] & \\ X^* \arrow[r, "f"] & Z \end{tikzcd}$$

*Proof.* (i) $f$ is continuous as we proved in the previous theorem. $f$ is clearly bijective by the definition of $X^*$. If $f$ is a homeomorphism then $f$ is a quotient map, so $g$ is also a quotient map. On the other hand, if $g$ is a quotient map, then $f$ is a quotient map from the previous theorem.

(ii) Suppose that $Z$ is Hausdorff. Then for any $X_1, X_2 \in X^*$, $f(X_1)$ and $f(X_2)$ have disjoint neighborhoods $U$ and $V$. As a result, $f^{-1}(U)$ and $f^{-1}(V)$ are disjoint neighborhoods of $X_1$ and $X_2$.

□

Here is an example showing that the product of two quotient maps need not to be a quotient map.

**Example 22.12.** *Let $X = \mathbb{R}$ and $X^*$ be the quotient space obtained from $X$ by identifying the subset $\mathbb{N}$ to a point $b$. Let $p : X \to X^*$ be the quotient map. Let $\mathbb{Q}$ be a subspace of $\mathbb{R}$; let $i : \mathbb{Q} \to \mathbb{Q}$ be the identity map. We show that*

$$p \times i : X \times \mathbb{Q} \to X^* \times \mathbb{Q}$$

*is not a quotient map.*

*For each $n$, let $c_n = \frac{\sqrt{2}}{n}$, and consider the lines in $\mathbb{R}^2$ with slopes 1 and -1 through the point $n \times c_n$. Let $U_n$ be the set of points of $X \times \mathbb{Q}$ that lie above or beneath both of the lines and between the lines $x = n - \frac{1}{4}$ and $x = n + \frac{1}{4}$. $U_n$ is open in $X \times \mathbb{Q}$. As a result, $U = \bigcup U_n$ is also open. $U$ is also saturated as it contains the inverse image of $b \times \mathbb{Q}$ under $p \times i$. Suppose $p \times i$ is a quotient map then $p(U) \ni b \times 0$ is a open set so there is a neighborhood of $b \times 0$, which we call $W \times I_\delta$ s.t. $I_\delta$ contains all the rationals with absolute value $< \delta$.*

*As a result, we have $p^{-1}(W) \times I_\delta \subset U$. Since $p^{-1}(W)$ is open in $X$ and contains $\mathbb{N}$, we have $(n-\epsilon, n+\epsilon) \subset W$ where $n$ is chosen large enough s.t. $c_n < \delta$.*

*As a result, $V = (n - \epsilon, n + \epsilon) \times I_\delta \subset U$ with $\epsilon < \frac{1}{4}$ (so the interval does not intersect the intervals of other $n$. However, it is clear that the block is not contained in $U_n$ which leads to contradiction.*

## Exercises

**Problem 2.** *(i) Let $p : X \to Y$ be a continuous map. Show that if there is a continuous map $f : Y \to X$ s.t. $p \circ f = 1_Y$, then $p$ is a quotient map.*

*(ii) If $A \subset X$, a **retraction** of $X$ onto $A$ is a continuous map $r : X \to A$ s.t. $r(a) = a$, $\forall a \in A$. Show that a retraction is a quotient map.*

*Proof.* (i) Notice $p$ has to be surjective. Suppose $U \subset Y$ with $p^{-1}(U)$ open in $X$. Then $f^{-1}(p^{-1}(U)) = 1_Y(U) = U$ is open in $Y$ since $f$ is continuous. Thus, $p$ is a quotient map.

(ii) The inclusion map $i : A \to X$ with $i(a) = a$ is continuous and $r \circ i = 1_A$, so $r$ is a quotient map by the above part.

□

**Problem 3.** *Let $\pi_1 : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be projection on the first coordinate. Let $A$ be the subspace of $\mathbb{R} \times \mathbb{R}$ consisting of all points $x \times y$ s.t. $x \geq 0$ or $y = 0$. Let $q : A \to \mathbb{R}$ be obtained by restricting $\pi_1$. Show that $q$ is a quotient map that is neither open nor closed.*

*Proof.*

Notice that $A = \overline{\mathbb{R}_+} \times \mathbb{R} \cup \mathbb{R} \times \{0\}$. Let $r : A \to \mathbb{R} \times \{0\}$ with $r(x \times y) = x \times 0$. $r$ is continuous as a restriction of a continuous function and is a quotient map as $r$ is a retraction. Additionally, $h : \mathbb{R} \times \{0\} \to \mathbb{R}$ is a homeomorphism and thus a quotient map. Hence, $q = h \circ r$ is a quotient map.

Notice that $\mathbb{R} \times \mathbb{R}_+ \cap A$ is open in $A$ but the image of it under $q$ is $\overline{R_+}$ which is not open in $\mathbb{R}$. Thus, $q$ is not an open map.

On the other hand, $C = \{x \times \tan x \mid \frac{\pi}{2} < x < \frac{3\pi}{2}\}$ is closed in $R$ and thus in $A$ but the map of it under $q$ is $(\frac{\pi}{2}, \frac{3\pi}{2})$ which is not closed in $\mathbb{R}$.

Notice that $C$ is closed because $\mathbb{R} \times \mathbb{R} - C$ is open as any point not on $C$ has a shortest distance to $C$, call it $\epsilon$. Then $B((x, y), \frac{\epsilon}{2})$ is a basis element lie in $\mathbb{R} \times \mathbb{R} - C$. □

**Problem 5.** *Let $p : X \to Y$ be an open map. Show that if $A$ is open in $X$, then the map $q : A \to p(A)$ obtained by restricting $p$ is an open map.*

*Proof.* Suppose $U \subset A$ is open, then $U = A \cap V$ is also open in $X$ so $p(U) = q(U)$ is open. Thus, $q$ is an open map. □

## 23 Connected Spaces

**Definition 23.1.** *Let $X$ be a connected space. A **separation** of $X$ is a pair of non-empty disjoint open sets $U$, $V$ s.t. $U \cup V = X$.*

*$X$ is called **connected** if there does not exist a separation of $X$.*

**Corollary 23.2.** *$X$ is connected iff there is no set $S \subset X$ s.t. $S$ is both open and closed.*

**Lemma 23.3.** *If $Y$ is a subspace of $X$, a separation of $Y$ is a pair of disjoint non-empty sets $A$ and $B$ whose union is $Y$, neither of which contains a limit point of the other. The space $Y$ is connected if there does not exist such a separation.*

*Proof.* Suppose that $A$ and $B$ form a separation of $Y$. As $A$ is closed $A = \bar{A} \cap Y$, implying that $\bar{A} \cap B = \varnothing$ Similar reasoning for $\bar{B}$.

On the other hand, suppose $A$ and $B$ are disjoint sets in $Y$ whose union is $Y$ s.t. neither of which contains a limit point of the other. Then we have that $\bar{A} \cap B = \varnothing$ so $\bar{A} \cap Y = \bar{A} \cap A = A$ and $\bar{B} \cap Y = B$. Thus, $A$ and $B$ are both closed and open in $Y$ so $Y$ is disconnected. □

**Lemma 23.4.** *If the sets $C$ and $D$ form a separation of $X$, and if $Y$ is a connected subspace of $X$, then $Y \subset C$ or $Y \subset D$.*

*Proof.* $C \cap Y$ and $D \cap Y$ are open in $Y$ and apparently they are disjoint. If $C \cap Y \neq \varnothing$ and $D \cap Y \neq \varnothing$ then they form a separation of $Y$, contradicting the fact that $Y$ is connected.

Suppose that $C \cap Y = \varnothing$ then $Y \subset D$. □

**Theorem 23.5.** *Suppose that $\{A_\alpha\}$ is a collection of connected subspaces of $X$. If $\bigcap A_\alpha \neq \varnothing$ then $\bigcup A_\alpha$ is connected.*

*Proof.* Let $p \in \bigcap A_\alpha$.

Suppose that there is a separation of $\bigcup A_\alpha$ by $U$ and $V$. Suppose that $p \in U$, then by the above lemma, $A_\alpha \subset U$ for all $\alpha$. Thus, $V$ is empty which leads to contradiction. As a result, $\bigcup A_\alpha$ is connected. □

**Theorem 23.6.** *Let $A$ be a connected subspace of $X$. If $A \subset B \subset \bar{A}$, then $B$ is also connected.*

*Proof.* Suppose that $B$ is not connected, then $B = U \cup V$ where $U$, $V$ are disjoint open set.

As a result of the lemma earlier, $A$ lies entirely in $U$ or $V$.

Suppose that $A \subset U$, then $\bar{A} \subset \bar{U}$. Since $\bar{U} \cap V = \varnothing$, $V$ is empty, which leads to contradiction. □

**Theorem 23.7.** *The image of a connected space under a continuous map is connected.*

*Proof.* Suppose $f : X \to Y$ is continuous and $f(X)$ is not connected while $X$ is connected. Then $f(X) = U \cup V$ where $U$, $V$ are disjoint open sets in $Y$. Then $f^{-1}(U) \cup f^{-1}(V)$ forms a separation of $X$ which contradicts the fact that $X$ is connected. Thus, $f(X)$ is connected. □

**Theorem 23.8.** *A finite Cartesian product of connected spaces is connected.*

*Proof.* Since we can prove the statement by induction, the theorem is equivalent to prove $X \times Y$ is connected if $X$ and $Y$ are.

Fix $(a \times b) \in X \times Y$. $X \times b$ is homeomorphic with $X$ so it is connected. For any $x \in X$, $x \times Y$ is homeomorphic with $Y$ so it is connected.

Notice that $(x \times b) \in X \times b \cup x \times Y$, so $X \times b \cup x \times Y$ is connected. Then, we have $X \times Y = \bigcup_{x \in X} X \times b \cup x \times Y$ is connected since $a \times b$ lies in all these spaces.

Thus, $X \times Y$ is connected. □

Notice that an infinite Cartesian product of connected spaces may or may not be connected.

**Example 23.9.** *Consider $\mathbb{R}^\omega$ in box topology. The product can be written as a disjoint union of the set of bounded sequences and the set of unbounded sequences.*

*Both of these are open, since if $\mathbf{a} \in \mathbb{R}^\omega$, then the set*

$$(a_1 - 1, a_1 + 1) \times ...$$

*is bounded or unbounded based on whether $\mathbf{a}$ is. Thus, $\mathbb{R}^\omega$ is not connected.*

**Example 23.10.** *On the other hand, consider the same product in the product topology. Let $\tilde{\mathbb{R}}^n$ denote the subspace of $\mathbb{R}^\omega$ where $x_i = 0$ if $i > n$. Clearly $\tilde{\mathbb{R}}^n$ is homeomorphic to $\mathbb{R}^n$ so it is connected. As a result, $\mathbb{R}^\infty$, as the union of all $\mathbb{R}^n$, with $\mathbf{0}$ in common, is connected.*

*We have proved that the closure of $\mathbb{R}^\infty = \mathbb{R}^\omega$ since any neighborhood containing $\mathbb{R}^\infty$ has all $U_n = \mathbb{R}$ except for finitely many $n$. Thus, $\mathbb{R}^\omega$ is connected as a result of **Theorem 23.6**.*

## Exercises

**Problem 5.** *A space is **totally disconnected** if its only connected subspaces are the one-point sets. Show that if $X$ has the discrete topology, then $X$ is totally disconnected. Does the converse hold?*

*Proof.*

Any space in discrete topology is apparently totally disconnected. On the other hand, $\mathbb{Q} \subset \mathbb{R}$ is also totally disconnected although the topology is the usual one. □

**Problem 6.** *Let $A \subset X$. Show that if $C$ is a connected subspace of $X$ that intersects both $A$ and $X - A$, then $C$ intersects* Bd $A$.

*Proof.*

Bd $A$ is the set of points $x$ s.t. for any neighborhood $U \ni x$, $U \cap A \neq \varnothing$, $U \cap X - A \neq \varnothing$. If $C \cap \text{Bd } A = \varnothing$, then $\forall c \in C$, $c$ has a neighborhood entirely lying in $U \cap A$ or $U \cap X - A$.

As a result, $A \cap C$ and $(X - A) \cap C$ are disjoint open sets that cover $C$, contradicting the fact that $C$ is connected. □

**Problem 7.** *Is $\mathbb{R}^\omega$ connected in the uniform topology?*

*Proof.* No. $A =$ the set of all bounded sequences and $B =$ the set of all unbounded sequences form a separation of $\mathbb{R}^\omega$. □

**Problem 9.** *Let $A \subset X$ and $B \subset Y$. If $X$ and $Y$ are connected, show that*

$$(X \times Y) - (A \times B)$$

*is connected.*

*Proof.* Let $c \times d \in (X - A) \times (Y - B)$. Similar to the idea for proving **Theorem 23.8**, we have $U = \bigcup\limits_{x \in X - A} (x \times Y) \cup (X \times d)$ connected, and $V = \bigcup\limits_{y \in Y - B} (c \times Y) \cup (X \times y)$ connected.

Therefore, $U \cup V = X \times Y - A \times B$ is connected. □

**Problem 10.** *Let $p : X \to Y$ be a quotient map. Show that if each set $p^{-1}(\{y]\})$ is connected and $Y$ is connected, then $X$ is connected.*

*Proof.* Suppose that $X = U \cup V$ is a separation of $X$. $p(U)$ and $p(V)$ are open and cover $Y$ since $p$ is a quotient map. WTS that $p(U)$ and $p(V)$ are disjoint.

Suppose that $y \in p(U) \cap p(V)$, then $\exists x_1 \in U$, $x_2 \in V$ s.t. $p(x_1) = p(x_2) = y$. Therefore, $x_1, x_2 \in p^{-1}(\{y\})$ which is connected.

However, by connectedness we should have that $p^{-1}(\{y\}) \subset U$ or $V$, leading to a contradiction.

Therefore, $p(U) \cap p(V) = \varnothing$ and form a separation of $Y$, leading to a contradiction. $\square$

## 24 Connected Subspaces of the Real Line

**Definition 24.1.** *A simply ordered set $L$ having more than one element is called a **linear continuum** if*

*(i) $L$ has the least upper bound property,*
*(ii) If $x < y$, $\exists z$ s.t. $x < z < y$.*

**Theorem 24.2.** *If $L$ is a linear continuum in the order topology, then $L$ is connected, and so are intervals and rays in $L$.*

*Proof.* Recall that in a convex subspace, the order topology is identical to the subspace topology. Thus, we directly prove that any convex subspace $Y$ of $L$ is connected.

Suppose that $Y = A \cup B$ with $A$, $B$ disjoint open subsets of $Y$. Since $A$, $B$ are nonempty, $\exists a \in A$ and $b \in B$, so $[a, b] \subset Y$. Thus, $[a, b] = A_0 \cup B_0$, where

$$A_0 = A \cap [a, b],\ B_0 = B \cap [a, b],$$

which form a separation of $[a, b]$.

Clearly $A_0$ is bounded as every element in $A_0$ is less than $b$. Let $c = \sup A_0$.

Suppose that $c \in A_0$. Since $A_0$ open in $[a, b]$, $\exists e > c s.t. [c, e) \subset A_0$, so $\exists c < p < e$ and $p \in A_0$ contradiction.

Suppose that $c \in B_0$. Since $B_0$ is open in $[a, b]$, $\exists d < c$ s.t. $(d, c] \subset B_0$, and $(d, c] \cup [c, b]$ does not intersect $A_0$ so $d$ is a smaller upper bound of $A_0$, which leads to contradiction.

Thus, any convex subspace of $Y$, including $L$ itself and any intervals or rays in $L$, is connected. □

**Corollary 24.3.** *The real line $\mathbb{R}$ is connected and so are intervals and rays in $\mathbb{R}$.*

**Theorem 24.4 (Intermediate Value Theorem).** *Let $f : X \to Y$ be a continuous map, where $X$ is a connected space and $Y$ is an ordered set with the order topology. If $a$ and $b$ are two points of $X$ and if $r$ is a point of $Y$ lying between $f(a)$ and $f(b)$, then $\exists c \in X$ s.t. $f(c) = r$.*

*Proof.* We know that f(X) is connected.

Suppose that $\nexists c \in X$ s.t. $f(c) = r$, then $(-\infty, r) \cup (r, \infty)$ is a separation of $f(X)$, contradicting the fact that it is connected. □

**Example 24.5.** *A typical example of a linear continuum different from $\mathbb{R}$ is the ordered square $I \times I$.*

*The first condition is quite trivial to check. The second one requires a case analysis about whether $\sup \pi_1(A) \in \pi_1(A)$, where $A$ is any bounded subset.*

*Details of the proof are not included here.*

**Definition 24.6.** *Given $x$, $y$ in a space $X$, a **path** in $X$ from $x$ to $y$ is a continuous map $f : [a, b] \to X$ s.t. $f(a) = x$ and $f(b) = y$.*

*A space $X$ is said to be **path connected** if every pair of points of $X$ can be joined by a path in $X$.*

**Corollary 24.7.** *A path connected space is always connected.*

*Proof.* If $X$ is a path connected space and $A$, $B$ forms a separation of $X$, then there will be no path from any point in $A$ to any point in $B$: the set $f([a, b])$ is connected, so it lies in either $A$ or $B$, so it cannot be a path from a point in $A$ to a point in $B$. □

**Example 24.8.** *Let the **unit disk** $D^n \subset \mathbb{R}^n$ by the equation*

$$D^n = \{\mathbf{x} \mid ||\mathbf{x}|| \leq 1\}.$$

*$D^n$ is path connected, as given any two points $\mathbf{x}$ and $\mathbf{y}$ in $D^n$, we always have $f(t) = (1-t)\mathbf{x} + t\mathbf{y}$ continuous, contained in $D^n$, with $f(0) = \mathbf{x}$ and $f(1) = \mathbf{y}$.*

*Similarly, every open ball and closed ball in $\mathbb{R}^n$ is path connected, and so is any convex subset.*

**Corollary 24.9.** *The image of a path connected space under a continuous map is path connected.*

*Proof.* Let $f : X \to Y$ be continuous, $f(x)$, $f(y)$ be two points in $Y$.

By path connectedness, there exists a path $\gamma_0 : xy$. Then $\gamma = f \circ \gamma_0$ will be a path from $f(x)$ to $f(y)$. Path connectedness follows. $\square$

**Example 24.10.** *Define the **unit square** $S^{n-1} \subset \mathbb{R}^n$ by the equation*

$$S^{n-1} = \{\mathbf{x} \mid ||\mathbf{x}|| = 1\}.$$

*If $n > 1$, it is path connected, since $g : \mathbb{R}^n - \{\mathbf{0}\} \to S^{n-1}$ where $g(\mathbf{x}) = \dfrac{\mathbf{x}}{||\mathbf{x}||}$ is continuous and surjective.*

*As the continuous image of a path connected space ($\mathbb{R}^n$ is path connected since we can always connect two points different from the origin by combining two straight lines), $S^{n-1}$ is path connected.*

**Example 24.11.** *Consider the ordered square $I_o^2$. It is connected as a linear continuum. However, it is not path connected.*

*Consider $p = 0 \times 0$ and $q = 1 \times 1$. Suppose $\exists f : [a, b] \to I_o^2$ s.t. $f(a) = p$ and $f(b) = q$. As the image of a connected space, $f([a, b])$ must contain every point $x \times y$ of $I_o^2$ by the intermediate value theorem.*

*Therefore, for each $x \in I$, the set*

$$U_x = f^{-1}(x \times (0, 1))$$

*is a nonempty subset of $[a, b]$. By continuity, it is open in $[a, b]$. For each $x \in I$, choose a rational number $q_x \in U_x$, then $x \to q_x$ is injective. However, this implies $I$ is countable which leads to contradiction.*

**Example 24.12.** *Let $S$ denote the following subset of the plane.*

$$S = \{x \times \sin(1/x) \mid 0 < x \leq 1\}.$$

*As the image of the connected set under a continuous map, $S$ is connected. Therefore, its closure $\bar{S}$ is also connected in $\mathbb{R}^2$. $\bar{S}$ is often called the **topologist's sine curve**. It is the union of $S$ and the set $0 \times [-1, 1]$.*

*Suppose there is a path $f : [a, c] \to \bar{S}$ from origin to some point in $S$. The set of $t$ s.t. $f(t) \in 0 \times [-1, 1]$ is closed, so it has a largest element $b$. Then $f : [b, c] \to \bar{S}$ is also a path that maps $b$ into the vertical interval $0 \times [-1, 1]$ and other points of $[b, c]$ to $S$.*

*Replace $[b, c]$ by $[0, 1]$. Let $f(t) = (x(t), y(t))$, then $x(0) = 0$, while $x(t) > 0$ and $y(t) = \sin(1/x(t))$ for $t > 0$. We want to show that there is a sequence of points $t_n \to 0$ s.t. $y(t_n) = (-1)^n$, then the sequence $y(t_n)$ does not converge, contradicting the continuity of $f$.*

*The construction is as follows. Given $n$, we choose $0 < u < x(1/n)$ s.t. $\sin(1/u) = (-1)^n$. By IVT, there exists $t_n$ s.t. $0 < t_n < 1/n$ with $x(t_n) = u$.*

*As a result, $\bar{S}$ is not path connected.*

## Exercises

**Problem 1.** *(i) Show that no two of the spaces $(0, 1)$, $(0, 1]$, and $[0, 1]$ are homeomorphic.*

*(ii) Suppose there exists imbeddings $f : X \to Y$ and $g : Y \to X$. Show by means of an example that $X$ and $Y$ need not to be homeomorphic.*

*(iii) Show $\mathbb{R}^n$ and $\mathbb{R}$ are not homeomorphic if $n > 1$.*

*Proof.* (i) Suppose that $(0, 1)$ is homeomorphic with $(0, 1]$, then $\exists f : (0, 1] \to (0, 1)$ continuous, then $f((0, 1]\backslash 1)$ should be connected but $(0, 1)\backslash f(1)$ is clearly not.

Similarly, $[0,1]$ with 0 and 1 removed is still connected, but $(0,1]$ with two points removed is clearly not.

Therefore, neither of the three spaces are homeomorphic.

(ii) Consider $f:(0,1)\to(0,1]$ with $f(x)=2x$ and $g:(0,1]\to(0,1)$ with $g(x)=\dfrac{x}{2}$, then $f$ and $g$ are imbeddings but $(0,1)$ and $(0,1]$ are not homeomorphic.

(iii) Similar to (i), suppose $f:\mathbb{R}^n\to\mathbb{R}$ is continuous, then $f(\mathbb{R}^n\backslash\mathbf{0})$ is connected but $\mathbb{R}\backslash f(\mathbf{0})$ is clearly not. □

**Problem 2.** *Let $f:S^1\to\mathbb{R}$ be a continuous map. Show there exists $x\in S^1$ s.t. $f(x)=f(-x)$.*

*Proof.* Consider $g(x)=f(x)-f(-x)$, let $x\in S^1$ s.t. $g(x)>0$, then $g(-x)<0$ and by the Intermediate Value Theorem we have $\exists y\in S^1$ with $g(y)=0$. □

**Problem 4.** *Show that a connected ordered set $X$ under the order topology is a linear continuum.*

*Proof.* Let $A$ be a subset of $X$ that has an upper bound. Suppose that $A$ does not have a least upper bound.

Consider

$$C=\bigcup_{a\in A}(-\infty,a) \text{ and } D=\bigcup_{b\in B}(b,\infty)$$

where $B$ is the set of all upper bounds of $A$. Apparently $C\cup D=X$ and they are disjoint open set, contradicting the fact that $X$ is connected.

Additionally, if there does not exist $c$ between $a<b$, then $(-\infty,b)$ and $(a,\infty)$ forms a separation of $X$, which is supposed to be connected.

Therefore, $X$ is a linear continuum. □

**Problem 5.** *Consider the following sets in the dictionary order. Which are linear continua?*

*(i)* $\mathbb{N}\times[0,1)$
*(ii)* $[0,1)\times\mathbb{N}$
*(iii)* $[0,1)\times[0,1]$
*(iv)* $[0,1]\times[0,1)$

*Proof.* (i) For any $x$ and $y$ in $\mathbb{N}\times[0,1)$ s.t. $x<y$, there apparently exists $z$ s.t. $x<z<y$.

Let $A$ be a upper bounded subset of $\mathbb{N}\times[0,1)$, let $x$ be the maximum value of the first coordinate in $A$, let $y=\sup\{y_i\mid x\times y_i\in A\cap x\times[0,1)\}$. If $y=1$, $(x+1)\times 0$ is the least upper bound for $A$ and if $y\neq 1$, then $x\times y$ is the least upper bound.

Thus, $\mathbb{N}\times[0,1)$ satisfies LUBP and is a linear continuum.

(ii) The set $A=\{0\times x\mid x\in\mathbb{N}$ is upper bounded but does not have a least upper bound.

(iii) For any upper bounded subset $A$, let $x=\sup\{x\mid x\times y\in A\}$ and $y=\sup\{y_i\mid x\times y_i\in A\cap x\times[0,1]\}$, then $x\times y$ is the least upper bound.

(iv) The set $A=\{0\times x\mid x\in[0,1)$ is upper bounded but does not have a LUB.

□

**Problem 7.** *(i) Let $X$ and $Y$ be ordered sets in the order topology. Show that if $f:X\to Y$ is order-preserving and surjective, then $f$ is a homeomorphism.*

*(ii) Let $X$ be the subspace $(\infty,-1)\cup[0,\infty)$ of $\mathbb{R}$. Show that the function $f:X\to\mathbb{R}$ defined by setting $f(x)=x+1$ if $x<-1$, and $f(x)=x$ if $x\geq 0$, is surjective and order-preserving. Are the two spaces homeomorphic?*

*Proof.* (i) $f$ is order-preserving so $f$ is injective, so $f$ is bijective.

$f$ is clearly continuous as inverse images of subbasis elements, such as $f^{-1}(-\infty, a) = (-\infty, f^{-1}(a))$, are open. Similarly, $f^{-1}$ is also continuous, so $f$ is a homeomorphism.

(ii) Clearly the two spaces are not homeomorphic by considering connectedness.

The reason is that the subspace topology on $X$ is not the same as the order topology. For those two to be the same, $X$ has to be convex.

□

**Problem 8.** *(i) Is a product of path-connected spaces necessarily path connected?*

*(ii) If $A \subset X$ and $A$ is path connected, is $\bar{A}$ necessarily path connected?*

*(iii) If $f : X \to Y$ is continuous and $X$ is path connected, is $f(X)$ necessarily path connected?*

*(iv) If $\{A_\alpha\}$ is a collection of path-connected subspaces of $X$ and if $\bigcap A_\alpha \neq \varnothing$, is $\bigcup A_\alpha$ necessarily path connected?*

*Proof.* (i) Yes. In the product topology a product of continuous functions is always a continuous function.

(ii) No. In the topologists' sine curve example, $S$ is path connected but $\bar{S}$ is not.

(iii) Yes. Composite of continuous functions is always continuous.

(iv) Yes. Take the gluing of the path from the common point to any $x_{\alpha_1}$ and $x_{\alpha_2}$ we get a path from $x_{\alpha_1}$ to $x_{\alpha_2}$, so the union is path connected.

□

**Problem 10.** *Show that if $U$ is an open connected subspace of $\mathbb{R}^2$, then $U$ is path connected.*

***Solution.*** Take $x \in U$, let the set of points that can be joined to $x$ by a path be denoted as $C_x$. We want to show that $C_x$ is both open and closed, leading to a contradiction.

For any $y \in C_x$, since $U$ is open, $\exists B(y, \epsilon) \subset U$. Clearly, for any $z \in B(y, \epsilon)$, $y$ can be connected to $z$ by a path in $B(y, \epsilon)$ (and thus in U). Since $y$ can be joined to $x$ by a path, $z$ can also be joined to $x$ by a path, so $B(y, \epsilon) \subset C_x$, so $C_x$ is open.

On the other hand, suppose $\exists y \in U$ s.t. $B(y, \epsilon) \cap C_x \neq \varnothing$ for any $\epsilon$. Then we have $z \in B(y, \epsilon) \cap C_x$ so $z$ can be joined to $x$ and $y$ by two paths. Connecting them we get a path joining $x$ to $y$. Thus, $y \in C_x$ so $C_x$ is also closed.

Since $U$ is connected, we have $U - C_x = \varnothing$ so $U$ is path connected. □

## 25 Components and Local Connectedness

**Definition 25.1.** *Given $X$ a topological space, define an equivalence relation on $X$ by setting $x \sim y$ if a connected subspace of $X$ contains both $x$ and $y$.*

*The equivalence classes are called the **components** of $X$.*

**Theorem 25.2.** *The components of $X$ are connected disjoint subspaces of $X$ whose union is $X$. Each nonempty connected subspace of $X$ intersects only one of them.*

*Proof.* Being equivalence classes, the components of $X$ are disjoint and their union is $X$. Suppose $A \cap X_1 \neq \varnothing$, $A \cap X_2 \neq \varnothing$ for some components $X_1$ and $X_2$ with $A$ connected, then there exists $x_1 \in X_1 \cap A$, $x_2 \in X_2 \cap A$.

Therefore, $x_1$ and $x_2$ are both in $A$ so they are in the same component, leading to a contradiction. □

**Corollary 25.3.** *Any component $C$ is connected.*

*Proof.* Let $x_0 \in C$, let $A_x$ be the connected subspace containing $x_0$ and $x$ for some $x \in C$. We then have $C = \bigcup\limits_{x \in C} A_x$.

Since the intersection of the $A_x$'s have $x_0$ in common, their union $C$ is connected. □

**Definition 25.4.** *We define another equivalence relation on the space $X$ by defining $x \sim y$ if there exists a path from $x$ to $y$.*

*The equivalence classes are called the **path components** of $X$.*

**Theorem 25.5.** *The path components of $X$ are path connected disjoint subspaces of $X$ whose union is $X$ s.t. each nonempty path connected subspace of $X$ intersects only one of them.*

*Proof.* The first part follows from the definition.

Suppose that there exists a path connected subspace $U$ of $X$ s.t. $U \cap X_1 \neq \varnothing$ and $U \cap X_2 \neq \varnothing$ for some path components $X_1$ and $X_2$. We then have $x_1 \in X_1 \cap U$ and $x_2 \in X_2 \cap U$ so $x_1$ and $x_2$ lie in the same path component, leading to a contradiction. □

**Corollary 25.6.** *Any component is closed in $X$. Any component is also open in $X$ if there are only finitely many components.*

*Proof.* This follows directly from the fact that the closure of a connected subspace is still connected. □

**Example 25.7.** *The path components of the subspace $\mathbb{Q} \subset \mathbb{R}$ are all singletons. None of them are open in $\mathbb{Q}$.*

**Example 25.8.** *The topologist's sine curve $\bar{S}$ has two path components, $\{0\} \times I$ and $S$. $S$ is open in $\bar{S}$ but not closed, and $\{0\} \times I$ is closed in $\bar{S}$ but not open.*

**Definition 25.9.** *A space $X$ is **locally connected at $x$** if for every neighborhood $U$ of $x$, there exists a connected neighborhood $V$ of $x$ contained in $U$. If $X$ is locally connected at all its points, it is said to be **locally connected**.*

*The definition of **locally path connected** is analogous.*

**Theorem 25.10.** *A space $X$ is locally connected iff for every open set $U$ of $X$, each component of $U$ is open in $X$.*

*Proof.* Suppose that $X$ is locally path connected. Let $x \in U \subset X$ with $U$ open. Suppose that $x \in U_i$ where $U_1$ is a component of $X$. We then have some connected $V \subset U$ that contains $x$.

Since $U_i$ is the component containing $x$, $V \subset U_i$, indicating that $U_i$ is open in $X$.

On the other hand, suppose that the latter condition holds, then for any $x \in X$ and $U$ neighborhood of $x$, the component of $U$ that contains $x$ is a connected neighborhood of $x$ contained in $U$. Locally connectedness therefore comes. □

Similarly, we have:

**Theorem 25.11.** *A space $X$ is locally path connected iff for every open set $U$ of $X$, each path component of $U$ is open in $X$.*

**Theorem 25.12.** *If $X$ is a topological space, each path component of $X$ lies in a component of $X$.*

*If $X$ is locally path connected, then the components and path components of $X$ are the same.*

*Proof.* The first part comes directly from the fact that the path components are connected (so any of them lie in a unique component).

Suppose that $X$ is locally path connected. Let $x \in C \cap P$ where $P$ is a path component and $C$ is a component. Let $Q$ be the union of path components that are not $P$ but intersect $C$. They all lie in $C$ by the former part.

We then have $C = P \cup Q$ and since $X$ is locally path connected, $P$ and $Q$ are both open and form a separation of $C$, leading to a contradiction. □

## Exercises

**Problem 3.** *Show that the ordered square is locally connected but not locally path connected. What are the path component of this space?*

*Proof.* Locally connectedness is clear. Notice that for any $0 < p < 1$, let $U$ be a neighborhood of $p \times 1$, then $U$ contains a path connected neighborhood, $V$, which has to contain some point $q \times 0$, where $q > p$.

Therefore, suppose $\gamma : [0, 1] \to I_o^2$ from $p \times 1$ to $q \times 0$, then by intermediate value theorem all the values in the form $r \times (0, 1)$ where $p < r < q$ is contained in $f([0, 1])$.

Therefore, $f^{-1}(r \times (0, 1))$ is open and we can form a bijection between $(p, q)$ and rationals on $[0, 1]$, which is clearly impossible.

Therefore, there is never a path from $(p, 1)$ to $(q, 0)$ if $p \neq q$, so the path components are the intervals $\{x\} \times [0, 1]$. □

**Problem 5.** *Let $X$ denote the rational points of the interval $[0, 1] \times 0$ of $\mathbb{R}^2$. Let $T$ denote the line segments from $p = (0, 1)$ to each point in $X$.*

*(i) Show that $T$ is path connected but locally connected only at $p$.*

*(ii) Find a subset of $\mathbb{R}^2$ that is path connected but locally connected at none of its points.*

*Proof.* (i) $T$ is path connected since any two points are both connected to $p$ by a path and by concatenating the two paths we get a new path between the two points.

$T$ is not locally connected at any other point because any neighborhood of points in $T \backslash \{p\}$ will be intersecting more than one line segment, so it cannot be connected.

(ii) N/A.

□

**Problem 8.** *Let $p : X \to Y$ be a quotient map. Show that if $X$ is locally connected, then $Y$ is locally connected.*

*Proof.* Let $U$ be an open set in $Y$ and $C$ be a component of $U$.Notice that the components of $p^{-1}(U)$ are connected and open, as $X$ is locally connected.

Let $U_\alpha$ be a component of $U$. Since $p(U_\alpha)$ is also connected, it must be contained in some component of $U$. In other words, either $p(U_\alpha) \subset C$ or $p(U_\alpha) \cap C = \varnothing$.

Therefore, either $U_\alpha \subset p^{-1}(C)$ or $U_\alpha \cap p^{-1}(C) = \varnothing$. Additionally, every point of $p^{-1}(C)$ must lie in a component of $p^{-1}(U)$, so $p^{-1}(C)$ is a union of some components of $p^{-1}(U)$.

Therefore, $p^{-1}(C)$ is a union of open sets and is therefore open, indicating that $C$ is open and $Y$ is locally connected.

$\square$

## 26 Compact Spaces

**Definition 26.1.** *A collection $\mathcal{A} = \{A_\alpha\}$ of subsets of a space $X$ is said to **cover** $X$, or to be the **covering** of $X$, if the union of the elements of $\mathcal{A}$ equals $X$.*

*It is called an **open covering** of $X$ if its elements are open subsets of $X$.*

**Definition 26.2.** *A space $X$ is said to be **compact** if every open cover of $X$ has a finite subcover that covers $X$.*

**Lemma 26.3.** *Let $Y$ be a subspace of $X$. Then $Y$ is compact iff every covering of $Y$ by sets open in $X$ contains a finite subcover covering $Y$.*

*Proof.* Suppose that $Y$ is compact and $\{A_\alpha\}$ is a covering of $Y$ by sets open in $X$, then

$$\{A_\alpha \cap Y\}$$

is also an open cover of $Y$ by sets open in $Y$. Thus, the open cover $\{A_\alpha\}$ has a finite subcover covering $Y$.

On the other hand, given an open cover of $Y$ by sets open in $Y$, $\{A'_\alpha\}$, for any $A'_\beta$ in the collection we have that $A'_\beta = A_\beta \cap Y$ for some $A_\beta$ open in $X$. By the assumption, $\{A_\alpha\}$ has a finite subcover covering $Y$ so $\{A'_\alpha\} = \{A_\alpha \cap Y\}$ also does, proving that $Y$ is compact. □

The lemma sheds light on an important fact: a space $X$ is compact iff $X$ is compact as a subspace of any space containing $X$.

**Theorem 26.4.** *Every closed subspace of a compact space is compact.*

*Proof.* Let $Y$ be a closed subspace of a compact space $X$. Given an open cover $\{A_\alpha\}$ of $Y$, consider the open cover of $X$:

$$\{A_\alpha\} \cup \{X - Y\}.$$

The open cover has a finite subcover covering $X$ due to compactness, so removing $X - Y$ from it we are guaranteed to get a finite subcover of $\{A_\alpha\}$ covering $Y$. □

**Theorem 26.5.** *Every compact subspace of a Hausdorff space is closed.*

*Proof.* Let $Y$ be a compact subspace of a Hausdorff space $X$. Suppose $x \notin Y$ is a limit point of $Y$. Since $X$ is Hausdorff, $\forall y \in Y$, we have neighborhoods $U_y$ and $V_y$ of $y$ and $x$ disjoint of each other.
Apparently, the collection

$$\{U_y \mid y \in Y\}$$

is an open cover of $Y$. Thus, there is a subcover

$$\{U_{y_i} \mid y_i \in Y, 1 \leq i \leq n\}$$

of $Y$ by compactness.

As a result, the set

$$V_{y_1} \cap ... \cap V_{y_k}$$

is a neighborhood of $x$ not intersecting $Y$, contradicting the fact that it is a limit point of $Y$. Therefore, $Y$ is closed. □

**Lemma 26.6.** *If $Y$ is a compact subspace of the Hausdorff space $X$ and $x_0 \notin Y$, then there exist disjoint open sets $U$ and $V$ of $X$ containing $x_0$ and $Y$, respectively.*

**Theorem 26.7.** *The image of a compact space under a continuous map is compact.*

*Proof.* Suppose $\{A_\alpha\}$ is an open cover of $f(X)$, then $\{f^{-1}(A_\alpha)\}$ is an open cover of $X$ so it contains a finite subcover

$$f^{-1}(A_{\alpha_1}),\ f^{-1}(A_{\alpha_2}),...,f^{-1}(A_{\alpha_k}),$$

covering $X$.

As a result, $A_{\alpha_1}$, ..., $A_{\alpha_k}$ is a finite subcover covering $f(X)$. Thus, $f(X)$ is compact. □

**Theorem 26.8.** *Let $f : X \to Y$ be a bijective continuous function. If $X$ is compact and $Y$ is Hausdorff, then $f$ is a homeomorphism.*

*Proof.* Let $C$ be closed in $X$, then by **Theorem 26.4**, $C$ is compact, by **Theorem 26.7**, $f(C)$ is compact, and by **Theorem 26.5**, $f(C)$ is closed in $Y$. Thus, $f^{-1}$ is continuous and therefore $f$ is a homeomorphism. □

**Theorem 26.9.** *The product of finitely many compact spaces is compact.*

*Proof.* We only need to prove that the product of two compact spaces is compact.

*Step 1.* Suppose that we are given spaces $X$ and $Y$ with $Y$ compact. Suppose that $x_0$ is a point of $X$ and $N$ is an open set of $X \times Y$ containing $x_0 \times Y$. We want to prove:

*There is a neighborhood $W$ of $x_0$ s.t. $W \times Y \subset N$.*

$W \times Y$ is called a ***tube*** about $x_0 \times Y$.

We now cover $x_0 \times Y$ by basis elements $U \times V \subset N$. As $x_0 \times Y$ is compact (homeomorphic to $Y$), we can cover $x_0 \times Y$ by finitely many basis elements

$$U_1 \times V_1,...,U_n \times V_n.$$

Consider the intersection $W = U_1 \cap ... \cap U_n$. As a finite intersection of open sets in $X$, it is open and contained in $\pi_1(N)$. Therefore, $\exists W \times Y \subset N$ with $W$ open and $x_0 \in W$.

*Step 2.* Now suppose that we are given an open cover $\mathcal{A}$ of $X \times Y$ where $X$ and $Y$ are compact.

For any $x_0 \in X$, $x_0 \times Y$ is compact and can therefore be covered by finitely many elements $A_{\alpha_1}, ..., A_{\alpha_n}$ in $\mathcal{A}$. Taking their union we have an open set $N$ containing $x_0 \times Y$. By Step 1, we know that $\exists$ neighborhood $W$ of $x_0$ s.t. $W \times Y \subset N$.

Apparently, by taking each $W$ for $x \in X$ we get an open cover of $X$. By compactness, we know there exists a finite collection

$$\{W_1, ..., W_k\}$$

covering $X$. Therefore, the union of

$$\{W_1 \times Y, ..., W_k \times Y\}$$

covers $X \times Y$ and each of them can be covered by finitely many elements in $\mathcal{A}$, giving the conclusion that there exists a finite subcover of $\mathcal{A}$ that covers $X \times Y$, so $X \times Y$ is compact. □

**Lemma 26.10** (The tube lemma)**.** *Consider the product space $X \times Y$, where $Y$ is compact. If $N$ is an open set of $X \times Y$ containing $x_0 \times Y$, then $N$ contains some tube $W \times Y$ about $x_0 \times Y$, where $W$ is a neighborhood of $x_0$.*

The tube lemma does not always hold if $Y$ is not compact.

**Example 26.11.** *Let $Y$ be the y-axis in $\mathbb{R}^2$. Then*

$$N = \{x \times y; |x| < 1/(y^2+1)\}$$

*contains* $0 \times \mathbb{R}$ *but does not contain any tube about it.*

**Definition 26.12.** *A collection* $\mathcal{C}$ *of subsets of* $X$ *is said to have the* ***finite intersection property*** *if for every finite subcollection*

$$\{C_1, ..., C_n\}$$

*of* $\mathcal{C}$*, the intersection* $C_1 \cap C_2 \cap ... \cap C_n$ *is not empty.*

**Theorem 26.13.** *Let* $X$ *be a topological space. Then* $X$ *is compact iff for every collection* $\mathcal{C}$ *of closed sets in* $X$ *having the finite intersection property, the intersection* $\bigcap_{C \in \mathcal{C}} C$ *of all the elements of* $\mathcal{C}$ *is not empty.*

*Proof.* Given a collection $\mathcal{A}$ of open sets of $X$, we can always generate a collection of closed set $\mathcal{C}$ by taking the complements of elements in $\mathcal{A}$. $X$ is compact is equivalent to saying if $\mathcal{A}$ is an open cover then it has a finite subcover covering $X$, which is equivalent to saying if $\bigcap_{C \in \mathcal{C}} C = \varnothing$ then $\exists C_1, ..., C_n \in C$ s.t, $C_1 \cap ... \cap C_n = \varnothing$.

Thus, if $\mathcal{C}$ is a collection of closed sets with the finite intersection property then the intersection $\bigcap_{C \in \mathcal{C}} C \neq \varnothing$, otherwise taking their complements we get an open cover $\mathcal{A}$ of $X$ without any finite subcover. □

## Exercises

**Problem 1.** *Show that if* $X$ *is compact Hausdorff under both* $\mathcal{T}$ *and* $\mathcal{T}'$*, then the two topologies are either equal or not comparable.*

***Solution.*** Notice that the identity map $i : (X, \mathcal{T}) \to (X, \mathcal{T}')$ is continuous if $\mathcal{T}' \subset \mathcal{T}$. Since $X$ is compact Hausdorff under both spaces, $i$ is a homeomorphism and the two topologies are therefore equal.

Otherwise the two topologies are not comparable and $i$ is therefore not continuous. □

**Problem 2.** *If* $\mathbb{R}$ *has the topology consisting of all sets* $A$ *s.t.* $\mathbb{R} - A$ *is countable or* $\mathbb{R}$*, then is* $[0, 1]$ *a compact subspace?*

***Solution.*** Let $B_n = \{\frac{1}{k} \mid k \geq n\}$ and $A_n = [0, 1] - B_n$, then $\{A_n\}$ is an open cover of $[0, 1]$ but apparently there's no finite subcover of it covering $[0, 1]$. □

**Problem 3.** *Show that every compact subspace of a metric space is bounded in that metric and is closed. Find a metric space in which not every closed bounded subspace is compact.*

***Solution.*** A compact subspace $A$ of a metric space is apparently closed as every metric space is Hausdorff. It is bounded since the open cover $B_d(x, n)$ where $x \in A$ and $n \in \mathbb{N}$ has a finite subcover, so $B_d(x, N)$ covers $A$ by taking $N$ to be the maximal $n$ among all balls of the subcover.

Consider an infinite set $X$ with the discrete metric. It is apparently closed and bounded since $X \subset B_d(x, 2)$ where $x$ is any point on the space. It is not compact, though, since the open cover $B_d(x, \frac{1}{2})$ where $x \in X$ does not have a finite subcover covering $X$. □

**Problem 7.** *Show that if* $Y$ *is compact, then the projection* $\pi_1 : X \times Y \to X$ *is a closed map.*

***Solution.*** Let $C$ be a closed subset of $X \times Y$, we want to show that $X - \pi_1(C)$ is open in $X$.

Let $x_0 \in X - \pi_1(C)$, then $x_0 \times Y \subset (X \times Y) \backslash C$, so by the tube lemma, $\exists W \ni x_0$, $W$ open in $X$, s.t. $W \times Y \subset (X \times Y) \backslash C$.

If $x \in W$, then $x \times y \in W \times Y$ for any $y \in Y$, so $x \times y \notin C$, so $W \subset X - \pi_1(C)$ and $X - \pi_1(C)$ is open. Thus, $\pi_1$ is a closed map. □

**Problem 9.** *Let $A$ and $B$ be subspaces of $X$ and $Y$, respectively; let $N$ be an open set in $X \times Y$ containing $A \times B$. If $A$ and $B$ are compact, then there exist open set $U$ and $V$ in $X$ and $Y$, respectively, s.t.*

$$A \times B \subset U \times V \subset N.$$

***Solution.*** For any $a \in A$, consider all the basis elements $W \times Z \subset N$ s.t. $a \in W$. Apparently we get an open cover of $a \times B$ and by compactness, we have $a \times B \subset W_1 \times Z_1 \cup ... \cup W_n \times Z_n$. Take

$$W_a = \bigcap W_i \text{ and } Z_a = \bigcup Z_i,$$

then we have $W_a$ and $Z_a$ open in $X$ and $Y$ and $a \times B \subset W_a \times Z_a \subset N$.

Consider the same process for each $a \in A$, we get an open cover $\{W_a \times Z_a\}_{\{a \in A\}}$ of $A \times B$. Since $A \times B$ is compact, there is a finite subcover covering $A \times B$; let it be $\{W_{a_1} \times Z_{a_1}, ..., W_{a_m} \times Z_{a_m}\}$, and take

$$U = \bigcup W_{a_i} \text{ and } V = \bigcap Z_{a_i},$$

then we have $A \times B \subset U \times V \subset N$. □

**Problem 10.** *Let $f_n : X \to \mathbb{R}$ be a sequence of continuous functions, with $f_n(x) \to f(x)$ for each $x \in X$. If $f$ is continuous, and if the sequence $f_n$ is monotone increasing, and if $X$ is compact, then the convergence is uniform.*

***Solution.*** Let $\epsilon$ be fixed. Since $f - f_n$ is continuous, $\forall x \in X$ we have a neighborhood $U_x$ about $x$ and some $N_x \in \mathbb{N}$ s.t. $f(z) - f_n(z) < \epsilon$ for $z \in U_x$, $n > N_x$. Since $X$ is compact, $X$ is covered by a finite subcover of it, namely

$$\{U_{x_1}, ..., U_{x_k}\}.$$

Let $N = \max(N_1, ..., N_k)$, then we have $f_n(x) - f(x) < \epsilon$ for any $n > N$, $x \in X$, thus proving the uniform continuity. □

## 27 Compact Subspaces of the Real Line

**Theorem 27.1.** *Let $X$ be a simply ordered set having the least upper bound property. In the order topology, each closed interval in $X$ is compact.*

*Proof.* Given $a$, $b \in X$. To show $[a, b]$ is compact we first try to show that for any $x \in [a, b]$ with $x \neq b$, $\exists y > x$ s.t. $[x, y]$ can be covered by at most two intervals of $A_\alpha$.

If $x$ has an immediate successor, let $y$ be it, then we have $[x, y]$ containing only two elements and clearly can be covered by at most 2 elements of $A_\alpha$. On the other hand, suppose that $x$ has no immediate successor, then choose an element $A$ from $A_\alpha$ that contains $x$. Since $x \neq b$, $\exists c > x$ s.t. $[x, c) \subset A$, so any element $y$ between $x$ and $c$ satisfies that $[x, y]$ can be covered by a single element of $A_\alpha$.

Let $x = a$ and $C$ be the set of all points $y > a$ s.t. $[a, y)$ can be covered by finitely many elements of $A_\alpha$. We know that $C$ is not empty by applying the earlier paragraph on $a$. Let $c$ be $\sup C$, then $a < c$.

Now we show that $c \in C$. Choose an element $A$ from $A_\alpha$ that covers $c$, then $\exists d < c$ s.t. $(d, c] \subset A$. If $c \notin C$, then $\exists z \in (d, c)$ and $z \in C$, since otherwise $d$ will be a smaller upper bound of $C$. Now we know $[a, z]$ can be covered by finitely many elements in $A_\alpha$ and if we add $A$ to that collection we are going to have a finite cover of $c$. Therefore, $c \in C$.

Now we suppose $c < b$, then $\exists y > c$ s.t. $[c, y]$ can be covered by at most 2 elements in $A_\alpha$. Therefore, $[a, y]$ can also be covered by finitely many elements so $y \in C$, contradicting that c $= \sup C$.

Therefore, $c = b$ so $[a, b]$ is compact. □

**Corollary 27.2.** *Every closed interval of $\mathbb{R}$ is compact.*

**Theorem 27.3.** *A subspace $A$ of $\mathbb{R}^n$ is compact iff it is closed and bounded in the Euclidean metric $d$ or the square metric $\rho$.*

*Proof.* As we have shown, the two metrics generate the same topology, it is suffice to prove the statement for $\rho$

Suppose that $A$ is compact, then $A$ is closed. Consider the collection of open balls $B_\rho(\mathbf{0}, n)$ where $n \in \mathbb{N}$, then the collection clearly covers $A$ and therefore has a finite subcover, which gives a value $N$ s.t. $A \subset B_\rho(\mathbf{0}, N)$. Therefore, $A$ is bounded.

On the other hand, suppose that $N$ is closed and bounded. Therefore there is some $N > 0$ s.t. $\rho(x, y) \leq N$ for any $x$, $y \in A$. Then choose any point $x_0 \in A$, let $b = \rho(x_0, \mathbf{0})$.

The triangular inequality tells that $\rho(x, \mathbf{0}) \leq N + b$ for any $x \in A$. Therefore, $A \subset [-N - b, N + b]^n$ so $A$ is a closed subset of a compact subspace, indicating that $A$ is compact. □

**Theorem 27.4** (Extreme Value Theorem)**.** *Let $f : X \to Y$ be continuous, where $Y$ is an ordered set in the order topology. If $X$ is compact, then there exists $c$, $d \in X$ s.t. $f(c) \leq f(x) \leq f(d)$ for any $x \in X$.*

*Proof.* Since $X$ compact and $f$ continuous, $f(X)$ is compact as well. If $f(X)$ has no maximal element, then we have

$$\{(-\infty, a) \mid a \in f(X)\}$$

covering $f(X)$ without a finite subcover, contradicting the claim that $f(X)$ is compact. Therefore, $\exists a = f(d)$ s.t. $f(x) \leq f(d)$ for any $x \in X$. The proof for the existence of $c$ is analogous. □

**Definition 27.5.** *Let $(X, d)$ be a metric space; let $A$ be a nonempty subset of $X$. For each $x \in X$, we define the* ***distance from x to A*** *by the equation*

$$d(x, A) = \inf\{d(x, a) \mid a \in A\}.$$

*We then have $d(x, A) - d(y, A) \leq d(x, y)$, so for fixed $A$, the function $d(x, A)$ is continuous. Explicitly, for any $\epsilon > 0$, $d(x, y) < \epsilon$ implies that $|d(x, A) - d(y, A)| < \epsilon$.*

*The* ***diameter*** *of a bounded subset $A$ and a metric space $(X, d)$ is*

$$\sup\{d(a_1, a_2) \mid a_1, a_2 \in A\}.$$

**Lemma 27.6.** *Let $\mathcal{A}$ be an open covering of a metric space $(X, d)$. If $X$ is compact, there is a $\delta > 0$ s.t. for each subset of $X$ with diameter less than $\delta$, there exists an element of $\mathcal{A}$ containing it.*

*$\delta$ is called the* ***Lebesgue number*** *for the covering $\mathcal{A}$.*

*Proof.* Choose a subcollection $\{A_1, ...A_n\}$ that covers $X$. For each $i$, let $C_i = X - A_i$, and define $f : X \to \mathbb{R}$ by letting $f(x)$ to be

$$f(x) = \frac{1}{n} \sum_{i=1}^{n} d(x, C_i).$$

Since $x \in A_i$ for some $i$ we have $f(x) > 0$. Since $f$ is continuous, it has a minimum value $\delta$.

Let $B$ be a subset of $X$ with diameter less than $\delta$. Choose an arbitrary point $x \in B$, then $B \subset B(x, \delta)$, and we also have

$$\delta \leq f(x) \leq d(x, C_m)$$

where $d(x, C_m)$ has the largest value among all $d(x, C_i)$.

Thus, the $\delta$-neighborhood is contained in $A_m$ since there is no point in $C_m$ with distance to $x$ less than $\delta$. Therefore, $B$ is contained in $A_m$. $\square$

**Definition 27.7.** *A function $f$ from the metric space $(X, d_X)$ to the metric space $(Y, d_Y)$ is* ***uniformly continuous*** *if given any $\epsilon > 0$, $\exists \delta > 0$ s.t. for every pair of $x_0$, $x_1 \in X$,*

$$d_X(x_0, x_1) < \delta \Rightarrow d_Y(f(x_0), f(x_1)) < \epsilon.$$

**Theorem 27.8.** *Let $f : X \to Y$ be a continuous map of the compact metric space $(X, d_X)$ to the metric space $(Y, d_Y)$. Then $f$ is uniformly continuous.*

*Proof.* Given $\epsilon > 0$, take the open covering of $Y$ by balls $B(y, \frac{\epsilon}{2})$ for every $y \in Y$. Let $A_\alpha$ be the open covering containing the inverse images of these balls.

Let $\delta$ be the Lebesgue number of this covering of $A$, then for any $x_0$, $x_1 \in X$ with $d(x_0, x_1) < \delta$, there exists some $f^{-1}(B(y, \frac{\epsilon}{2}))$ s.t. $x_0$, $x_1 \in f^{-1}(B(y, \frac{\epsilon}{2}))$.

Therefore, $d(f(x_0), f(x_1)) < \epsilon$ if $d(x_0, x_1) < \delta$. Thus, $f$ is uniformly continuous. $\square$

**Definition 27.9.** *If $X$ is a space, a point $x \in X$ is called an* ***isolated point*** *if $\{x\}$ is open in $X$.*

**Theorem 27.10.** *Let $X$ be a nonempty compact Hausdorff space. If $X$ has no isolated point then $X$ is uncountable.*

*Proof.* We first show that given any nonempty open set $U$ of $X$ and any $x \in X$, $\exists V$ nonempty and open s.t. $V \subset U$ and $x \notin \bar{V}$.

Choose a point $y \in U$ s.t. $y \neq x$ (which is possible because if $x \notin U$ then $\exists y \in U$ and if $x \in U$ then $\exists y \in U$ or otherwise $x$ is an isolated point). Choose $W_1$, $W_2$ disjoint neighborhood of $x$ and $y$, then $V = W_2 \cap U$ is the desired open set. $x \notin \bar{V}$, because $W_1$ is open so it does not contain any limit point of $W_2$.

Now we show that given $f : \mathbb{N} \to X$, $f$ is not surjective, indicating that $X$ is not countable. Let $x_n = f(n)$, then we have a nonempty open set $V_1 \in X$ s.t. $x_1 \notin \bar{V_1}$. Then we choose $\bar{V_n}$ s.t. $V_n \subset V_{n-1}$ open and nonempty, with $x_n \notin \bar{V_n}$. Consider the nested sequence

$$\bar{V_1} \supset ... \supset \bar{V_n} \supset ...$$

As $X$ is compact, $\exists x \in \bigcap \bar{V}_n$, indicating that $f$ is not surjective since $x \neq x_n$ for any $n$, thus proving that $X$ is uncountable. $\square$

**Corollary 27.11.** *Every closed interval in $\mathbb{R}$ is uncountable.*

## Exercises

**Problem 1.** *Prove that if $X$ is an ordered set in which every closed interval is compact, then $X$ has the least upper bound property.*

*Proof.* Let $A$ be a bounded subset of $X$. Notice that $\bar{A}$ is closed and is thus compact.

Thus, $i : \bar{A} \to X$ has a maximum value, call it $b$, s.t. $a \leq b$ for any $a \in \bar{A}$. We want to show that $b$ is the least upper bound of $A$.

Suppose that $\exists c < b$ s.t. $a \leq c$ for any $a \in A$. Let $\epsilon = d(c, b)$, then $(b - \frac{\epsilon}{2}, b + \frac{\epsilon}{2}) \cap A = \varnothing$, contradicting to the fact that $b \in \bar{A}$.

Thus, there does not exist such $c$ and $b$ is indeed the least upper bound of $A$. $\square$

**Problem 4.** *Show that a connected metric space having more than one point is uncountable.*

*Proof.* We have proved that $d : X \times X \to \mathbb{R}$ is continuous in **Exercise 20.3**. Thus, $d_x : X \to \mathbb{R}$ with $d_x(y) = d(x, y)$ is continuous too.

Since $X$ has more than one point, $\exists y \in X$ s.t. $x \neq y$. Let $d_x(y) = \delta$. Thus, $d_x(X)$ as a connected subspace of $\mathbb{R}$ contains $[0, \delta)$, so $X$ has to be uncountable. $\square$

**Problem 5.** *Let $X$ be a compact Hausdorff space; let $\{A_n\}$ be a countable collection of closed sets of $X$. Show that if each set $A_n$ has empty interior in $X$, then the union $\bigcup A_n$ has empty interior in $X$.*

***Solution***. Let $U$ be an arbitrary open set in $X$. We know that $U \not\subset A_1$ since $A_1$ has an empty interior.

As a result, $\exists x \in U \backslash A_1$ which indicates $x \notin A_1 \cup (X \backslash U)$. Since $A_1 \cup (X \backslash U)$ is closed in $X$ and $X$ is compact, $A_1 \cup (X \backslash U)$ is compact, indicating that there is a neighborhood $V_1$ of $x$ that is disjoint from an open set $U_1$ in $A_1 \cup (X \backslash U)$. As a result, $x \in V_1 \subset \bar{V}_1 \subset X \backslash (A_1 \cup (X \backslash U)) = U \backslash A_1$.

Now, $V_1$ is an open set so it is not contained in $A_2$, so there exists $V_2$ open in $A_2$ s.t. $\bar{V}_2 \subset V_1 \backslash A_2$. Repeating the process, we get a nested sequence

$$\bar{V}_1 \supset \bar{V}_2 \supset \ldots.$$

Since $X$ is compact, $\bigcap \bar{V}_i$ is not empty, indicating that there is a $x$ lying in $\bar{V}_i \subset U$ for any $i$ so it is not in any of the $A_i$'s. Thus, $U$ has a point not in $\bigcup A_n$, indicating that $\bigcup A_n$ has empty interior. $\square$

## 28 Limit Point Compactness

**Definition 28.1.** *A space $X$ is called **limit point compact** if every infinite subset of $X$ has a limit point.*

**Theorem 28.2.** *Compactness implies limit point compactness, but the converse does not hold.*

*Proof.* We prove the contrapositive: if $A$ has no limit point in a compact space $X$, then $A$ must be finite.

Since $A$ has no limit point, $A$ is closed. For any $x \in A$, since $x$ is not a limit point of $A$, there exists a neighborhood $U_x$ of $x$ such that $U_x \cap A = \{x\}$.

Therefore, $\{U_x\} \cup \{X \backslash A\}$ form an open cover of $X$, so it has a finite subcover.

Therefore, $\{U_x\}$ has a finite subcover that covers $A$, so $A$ has to be finite. □

Here is an example of a space $X$ that is limit point compact but not compact.

**Example 28.3.** *Let $Y$ consists of two points; give $Y$ the topology consisting of $Y$ and $\varnothing$.*

*Let $X = \mathbb{N} \times Y$, then $X$ is limit point compact. For if $(x, a) \in M \subset \mathbb{N} \times Y$, we have $(x, b)$ a limit point of $M$, since any neighborhood of $(x, b)$ contains $(x, a)$.*

*However, $X$ is not compact since the open cover $\{U_x = \{x\} \times Y, x \in \mathbb{N}\}$ does not have a finite subcover covering $X$.*

**Definition 28.4.** *Let $X$ be a topological space. If $(x_n)$ is a sequence of points of $X$ with*

$$n_1 < ... < n_i < ...$$

*an increasing sequence of positive integers, then the sequence $(y_i) = (x_{n_i})$ is called a **subsequence** of $(x_n)$. $X$ is called **sequentially compact** if every sequence of points of $X$ has a convergent subsequence.*

**Theorem 28.5.** *Let $X$ be a metrizable space. Then the followings are equivalent:*

*(i) $X$ is compact.*

*(ii) $X$ is limit point compact.*

*(iii) $X$ is sequentially compact.*

*Proof.* (i)$\Rightarrow$ (ii) is shown.

Suppose that $X$ is limit point compact. Let $(x_n)$ be a sequence of points in $X$. If $(x_n)$ only has a finite range, then there is at least one value $x$ appearing infinitely many times, so $x$ is a limit point of $(x_n)$.

On the other hand, if $(x_n)$ has an infinite range, then it has a limit point since $X$ is limit point compact. Call the limit point $x$

We can then take a subsequence $(x_{y_i})$ where $d(x_{y_k}, x) < \frac{1}{k}$ and $y_k < y_{k+1}$ for any $i$. Clearly, we have $(x_{y_i})$ a subsequence converging to $x$, so $X$ is sequentially compact.

Now suppose that $X$ is sequentially compact. We first show that the Lebesgue number lemma holds for $X$. Let $\mathcal{A}$ be an open cover of $X$. Assume that there is no $\delta > 0$ such that every set of diameter less than $\delta$ has an element of $\mathcal{A}$ covering it, then for any $n \in \mathbb{N}$, we can find $C_n \subset X$ not contained in any element of $\mathcal{A}$ with diameter less than $\frac{1}{n}$.

Let $x_n \in C_n$, then the sequence $(x_n)$ has a subsequence $x_{n_i}$ that converges and has a limit point $a$. Since $a \in A_\alpha$ for some $\alpha$, there exists $\epsilon > 0$ s.t. $B(a, \epsilon) \subset A_\alpha$.

Let $n_i$ be large enough such that $\frac{1}{n_i} < \frac{\epsilon}{2}$, then $C_{n_i}$ is contained in the neighborhood of $x_{n_i}$ with diameter $\frac{\epsilon}{2}$. Suppose that $i$ is also large enough such that $d(x_{n_i}, a) < \dfrac{\epsilon}{2}$, then we have $C_{n_i} \subset B(x_{n_i}, \frac{\epsilon}{2}) \subset B(a, \epsilon) \subset A_\alpha$, which leads to a contradiction.

Next, we show that if $X$ is sequentially compact, then there exists a finite open covering of $X$ by $\epsilon$-balls for any given $\epsilon > 0$. Suppose not, then we choose $x_1 \in X$ and $x_n \in X \backslash \bigcup B(x_i, \epsilon)$, then $(x_n)$ is a sequence and has a converging subsequence. However, $d(x_i, x_j) \geq \epsilon$, so $(x_n)$ cannot have a converging subsequence, which leads to contradiction.

Finally, we show that $X$ is compact. Let $\mathcal{A}$ be an open covering of $X$ and let $\delta$ be the Lebesgue number of $X$ relative to $\mathcal{A}$. Let $\epsilon = \frac{\delta}{3}$, then $X$ is covered by a finite collection of $\epsilon$-balls.

Since these balls have diameter at most $\frac{2\delta}{3}$, each of them is contained in an element of $\mathcal{A}$. These elements form a finite subcover that covers $X$, so $X$ is compact. □

## Exercises

**Problem 1.** *Give* $[0,1]^\omega$ *the uniform topology. Find an infinite subset of this space without a limit point.*

***Solution.*** Consider $(x_i) = (0, 0, .., 1, 0, ...)$ where the $i$-th index is 1 and the others are all 0s. Clearly it has no limit point, as for any point $x \in [0,1]^\omega$, $B(x, \dfrac{1}{2})$ contains at most one point of the given set. □

**Problem 2.** *Show that* $[0,1]$ *is not limit point compact as a subspace of* $\mathbb{R}_l$.

***Solution.*** Let $A = \{1 - \dfrac{1}{n} \mid n \in \mathbb{N}\}$, then $A$ has no limit point.

In particular, 1 is not a limit point because $\{1\} = [0,1] \cap [1,2)$ is open under the subspace topology of $[0,1]$, and it contains no point of $A$. □

**Problem 3.** *(i) If* $f : X \to Y$ *is continuous, then is* $f(X)$ *necessarily limit point compact?*
*(ii) If* $A$ *is a closed subset of* $X$*, then is* $A$ *limit point compact?*

***Solution.*** (i) No. Consider **Example 28.3**. $X$ is limit point compact and the projection map $f : X \to \mathbb{N}$ is continuous, but $\mathbb{N}$ is not limit point compact.

(ii) Yes. Let $Y \subset A$ be an infinite subset, then $Y$ has a limit point in $X$ because $X$ is limit point compact. As $A$ is closed, that limit point lies in $A$, so $Y$ has a limit point in $A$ and $A$ is limit point compact.

□

**Problem 5.** *Show that* $X$ *is countably compact iff every nested sequence* $C_1 \supset ...$ *of closed nonempty sets of* $X$ *has a nonempty intersection.*

***Solution.*** Suppose $X$ is countably compact. If $\bigcap C_i = \varnothing$ then $\{X - C_i\}$ covers $X$, so there is an $N \in \mathbb{N}$ such that $C_N = \varnothing$, since $\{X - C_i\}$ needs to have a finite subcollection covering $A$, contradicting the fact that $C_N \neq \varnothing$. Therefore, $\bigcap C_i \neq \varnothing$.

Now, suppose that $\bigcap C_i \neq \varnothing$ for any such nested sequence $(C_i)$. Suppose that $\{A_n\}$ is an open cover of $X$, then we have $C_i = X - \bigcup_{j=1}^{i} A_j$ a nested sequence of closed sets.

Since $\{A_n\}$ is a countable open cover and that nested sequence of nonempty closed sets has a nonempty intersection, we have $C_N = \varnothing$ for some $N \in \mathbb{N}$.

Therefore, $\{A_N\}$ covers $X$ and $X$ is therefore countably compact. □

## 29 Local Compactness

**Definition 29.1.** *A space $X$ is **locally compact at** $x$ if there is some compact subspace $C$ of $X$ that contains a neighborhood of $x$. $X$ is **locally compact** if $X$ is locally compact at each of its points.*

**Example 29.2.** *$\mathbb{R}$ is locally compact since any neighborhood $(a, b)$ for $a < x < b$ is contained in $[a, b]$, which is compact.*

*On the other hand, $\mathbb{Q}$ is not locally compact. We will show this by showing that $[-\epsilon, \epsilon] \cap \mathbb{Q}$ is not compact for any $\epsilon > 0$, as this implies that $\mathbb{Q}$ is not locally compact at 0.*

*Now, if $[-\epsilon, \epsilon] \cap \mathbb{Q}$ is compact, then it is sequentially compact because $\mathbb{Q}$ is metrizable. However, if $(a_n) \subset [-\epsilon, \epsilon] \cap \mathbb{Q}$ is a sequence converging to some $a \in \mathbb{R}\backslash\mathbb{Q}$, then $(a_n)$ has no converging subsequence in $[-\epsilon, \epsilon] \cap \mathbb{Q}$, so $[-\epsilon, \epsilon] \cap \mathbb{Q}$ is not sequentially compact and hence not compact.*

*$\mathbb{R}^n$ is locally compact for the same reason of $\mathbb{R}$ being locally compact.*

*$\mathbb{R}^\omega$ is not locally compact. Notice that any of its basis elements $B = (a_1, b_1) \times ... \times (a_n, b_n) \times \mathbb{R} \times ...$ contains infinitely many $\mathbb{R}$'s as components. Therefore, suppose that $B \subset C$ for some compact $C \subset \mathbb{R}^\omega$, then $C$ is closed because $\mathbb{R}^\omega$ is Hausdorff. Therefore, $\bar{B} \subset C$ and as a closed subset of a compact set, $\bar{B} = [a_1, b_1] \times ... \times [a_n, b_n] \times \mathbb{R} \times ...$ is compact, which clearly is not true as $\mathbb{R}$ is not compact.*

**Theorem 29.3.** *Let $X$ be a space. Then $X$ is locally compact Hausdorff iff there exists a space $Y$ satisfying the following conditions:*

*(a) $X$ is a subspace of $Y$.*
*(b) The set $Y - X$ consists of a single point.*
*(c) $Y$ is compact Hausdorff.*

*Moreover, if $Y$ and $Y'$ are two spaces satisfying these, then there is a homeomorphism of $Y$ with $Y'$ that equals the identity map on $X$.*

*Proof.* We first verify uniqueness. Let $Y$ and $Y'$ be two spaces satisfying these conditions. Let $h : Y \to Y'$ with $h(x) = x$ if $x \in X$ and $h(p) = q$ otherwise. We want to show that if $U$ open in $Y$ then $h(U)$ is open in $Y'$.

If $U$ does not contain $p$, then $h(U) = U$. Since $U$ open in $Y$ and $X$ subspace of $Y$, $U$ is open in $X$, indicating that $U$ open in $Y'$.

If $U$ contains $p$, then $C = Y - U$ is closed and is thus compact as $Y$ is compact. Since $C$ is contained in $X$, $C$ is compact in $X$ and thus compact in $Y'$. Because $Y'$ is Hausdorff, $C$ is closed in $Y'$. Therefore, $h(U) = Y' - C$ is open.

Therefore, $h$ is a homeomorphism by some symmetric argument, and $Y \cong Y'$.

Now, suppose that $X$ is locally compact Hausdorff and we construct $Y$. Let $Y = \{\infty\} \cup X$. We define the topology on $Y$ as $U$ is open in $Y$ if either $U \subset X$ and open in $X$ or $U = Y - C$ where $C$ is compact in $X$. This is a natural definition, because we need the complement of closed subsets of $Y$ to be open, and closed subsets that do not contain $\infty$ are equivalent to compact subsets in $X$.

To verify that it is actually a topology, we take a look at the possible finite intersections of open sets, which are

$$U_1 \cap U_2$$
$$(Y - C_1) \cap (Y - C_2) = Y - (C_1 \cup C_2)$$
$$U_1 \cap (Y - C_1) = U_1 \cap (X - C_1),$$

all of which are open in our topology. Similarly, we check arbitrary unions of open sets, which are

$$\bigcup U_\alpha$$
$$\bigcup (Y - C_\beta) = Y - \bigcap C_\beta$$
$$\bigcup U_\alpha \cup \bigcup (Y - C_\beta) = U \cup (Y - C) = Y - (C - U),$$

all of which are open in our topology (the last one is open because $C - U$ is a closed subspace of $C$ and therefore compact).

We then need to show that $X$ is a subspace of $Y$. If $U$ is open in $Y$ and $\infty \notin X$, then $U \cap X = U$ is open in $X$. If $\infty \in U$, then $C = Y - U$ is compact. Therefore, $(Y - C) \cap X = X - C$ is open in $X$. On the other hand, a subset $U$ open in $X$ is always open in $Y$ by our definition.

Now we want to show that $Y$ is compact. Suppose $\mathcal{A}$ is an open cover of $Y$, then one of them, call it $U = Y - C$, has to contain the point $\infty$. Also, there is a collection of finitely many elements in $\mathcal{A}$ that covers $C$. Adding $U$ to the collection we get a finite cover of $Y$. Therefore $Y$ is compact.

Now we want to show that $Y$ is Hausdorff. If $x, y \in X$, then apparently there are neighborhoods $U$ and $V$ of them that are disjoint, which are also open and disjoint in $Y$. On the other hand, for any $x \in X$ and $\infty$, since $X$ is locally compact, there exists a compact subset $C$ containing a neighborhood $U$ of $x$. $Y - C$ is then a neighborhood of $\infty$ disjoint from $U$. Therefore, $Y$ is Hausdorff.

We now prove the converse. As a subspace of $Y$, $X$ is clearly Hausdorff. For any $x \in X$, let $U$ and $V$ be disjoint neighborhoods of $x$ and $\infty$, then we have $U \subset Y - V$ which is closed in $Y$ and thus compact. Therefore, $Y - V$ is compact in $X$ and $X$ is locally compact. □

**Remark 29.4.** *If $X$ is compact, then $Y$ is just $X$ adjoined with an isolated point. If $X$ is not compact, on the other hand, then $\infty \in Y - X$ is a limit point of $X$, so $\bar{X} = Y$.*

**Definition 29.5.** *If $Y$ is a compact Hausdorff space and $X$ is a proper subspace of $Y$ whose closure equals $Y$, then $Y$ is said to be a* ***compactification*** *of $X$. If $Y - X$ consists of one point, then $Y$ is called the* ***one-point compactification*** *of $X$.*

**Corollary 29.6.** *$X$ is locally compact Hausdorff $\Leftrightarrow$ $X$ compact or has a one-point compactification.*

**Example 29.7.** *The one-point compactification of $\mathbb{R}$ is the circle. Similarly, $S^2$ is the one-point compactification of $\mathbb{R}^2$.*

*The idea of the proof is to prove that $S^n$ without the north-pole $(0, ..., 0, 1)$ is homeomorphic to $\mathbb{R}^n$. To show this, consider the function*

$$p : (x, t) \to \frac{1}{1-t} x$$

*with $x \in \mathbb{R}^n$, which is well-defined since $t = 0$ iff $x = (0, ..., 0)$ which is already excluded from $S^n$. The inverse of the function is*

$$p^{-1}(y) = \frac{1}{||y||^2 + 1}(2y, ||y||^2 - 1),$$

*which maps $\mathbb{R}^n$ to $S^n \backslash N$ as can be directly confirmed by algebraic computation.*

**Theorem 29.8.** *Let $X$ be a Hausdorff space. Then $X$ is locally compact iff given $x \in X$ and a neighborhood $U$ of $x$, there is a neighborhood $V$ of $x$ s.t. $\bar{V}$ is compact and $\bar{V} \subset U$.*

*Proof.* Clearly the formulation implies local compactness as $C = \bar{V}$ is compact and contains a neighborhood of $x$, namely $V$.

To prove the converse, suppose that $X$ is locally compact. Let $x \in X$ and $U$ a neighborhood of $x$. Take the one-point compactification $Y$ of $X$.

$U$ is then open in $Y$ and $Y - U$ is closed and thus compact in $Y$. Therefore, $\exists V$ and $W$ disjoint open sets containing $x$ and $Y - U$. Additionally, $\bar{V}$ is compact in $Y$, as $\bar{V}$ is closed and $Y$ is compact. Therefore, $\bar{V}$ is compact. We now want to show that $\bar{V} \cap (Y - U) = \varnothing$, which is true because $\bar{V} \cap W = \varnothing$, as $W$ is open. Therefore, $\bar{V} \subset U$ and the statement is hence satisfied. □

**Corollary 29.9.** *Let $X$ be locally compact Hausdorff; let $A$ be a subspace of $X$. If $A$ is closed or open in $X$, then $A$ is locally compact.*

*Proof.* Suppose that $A$ is closed in $X$. Given $x \in A$, let $C$ be the compact subspace of $X$ containing a neighborhood $U$ of $x$ in $X$. Then $C \cap A$ is closed in $C$ and thus compact. Since it contains $U \cap A$, we have $A$ locally compact.

Suppose $A$ is open in $X$. Given $x \in X$, choose a neighborhood $V$ of $x$ in $X$ such that $\bar{V}$ is compact and $\bar{V} \subset A$. Then $C = \bar{V}$ is a compact subspace of $A$ containing the neighborhood $V$ of $x$ in $A$, so $A$ is locally compact. □

**Corollary 29.10.** *A space $X$ is homemomorphic to a proper open subspace of a compact Hausdorff space iff $X$ is locally compact Hausdorff.*

*Proof.* Let $Y$ be compact Hausdorff and $h : X \to Y$ a continuous map such that $h : X \to h(X)$ is a homeomorphism. Therefore, let $y \in Y \backslash h(X)$, then $Y - \{y\}$ is locally compact Hausdorff, and if $h(X)$ is open, then $X$ can be seen as an open subspace of $Y - \{y\}$, so it is locally compact Hausdorff.

On the other hand, if $X$ is locally compact Hausdorff, then we can view it as an open subspace of its one-point compactification, as by our definition $X$ is open in $Y$ since it is open in $X$. □

## Exercises

**Problem 2.** *Let $\{X_\alpha\}$ be an indexed family of nonempty basis. Show that if $\prod X_\alpha$ is locally compact, then each $X_\alpha$ is locally compact and $X_\alpha$ is compact for all but finitely many $\alpha$.*

***Solution.*** Given any $p \in X_\alpha$, let $x \in \prod X_\beta$ with $x_\alpha = p$.

We then have some neighborhood of $x$ and some compact subspace $C$ such that $U \subset C$. Therefore, there exists a basis element $\prod V_\beta \subset U$. Also, $\pi_\alpha : \prod X_\beta \to X_\alpha$ is continuous so $\pi_\alpha(C)$ is compact in $X_\alpha$.

Therefore, $p = x_\alpha \in V_\alpha \subset \pi_\alpha(C)$ so $X_\alpha$ is locally compact.

Also, notice that all but finitely many $V_\beta$'s are $X_\beta$'s, so $\pi_\beta(C) = X_\beta$ for all but finitely many $\beta$, so $X_\beta$ is compact for all but finitely many $\beta$. □

**Problem 3.** *Let $X$ be a locally compact space. If $f : X \to Y$ is continuous, does it follow that $f(X)$ is locally compact? What if $f$ is both continuous and open?*

***Solution.*** The former is not necessarily true, as $f$ preserves compact but not open subsets. For example, $i : \mathbb{Q}_d \to \mathbb{Q}$ is continuous, and $\mathbb{Q}_d$ is locally compact as it has the discrete topology so every point itself is a neighborhood. However, $\mathbb{Q}$ is not locally compact as we proved.

The latter is true if $f$ is surjective, as for any $y \in Y$, we can pick $x \in X$ and correspondingly $U$ and $C$ open and compact subsets, respectively, such that $x \in U \subset C$. Therefore, $f(U)$ and $f(C)$ are open and compact, respectively, and $y \in f(U) \subset f(C)$, so $Y$ is locally compact. □

**Problem 4.** *Show that $[0,1]^\omega$ is not locally compact in the uniform topology.*

*Proof.* Let $C$ be a compact subspace of 0 with $B(0,\epsilon) \subset C$. We then have $\overline{B(0,\epsilon)} = [0,\epsilon]^\omega \subset C$, so $[0,\epsilon]^\omega$ is compact.

However, this cannot be true as $[0,1]^\omega$ is not compact because it is not limit point compact, as we proved in the exercises of the last section. □

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